

# Joint Phase Reconstruction and Cell Segmentation from Raw Holograms for Per-Cell Dry-Mass Measurement

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## Abstract

Per-cell dry mass in quantitative phase imaging is the phase integrated over a segmented cell, so its accuracy depends on the reconstructed phase and on the cell boundary together. We trained a network of 9.6 M parameters, a MobileNetV2 encoder with two U-Net-style decoders, that maps a raw digital hologram to a quantitative phase map and a cell segmentation, and we scored every prediction through a fixed per-cell measurement chain as well as with phase and segmentation metrics. Segmentation labels were derived from the reference phase. On the off-axis test set (113 fields, 3186 reference cells, three training seeds), the dry-mass mean absolute percentage error (MAPE) over matched cells was 0.1742, against 0.2203 for the tested classical reconstruction pipeline, at similar Dice (0.8337 and 0.8253) and projected-area MAPE (0.1568 and 0.1578). On the same test set, detection recall was 0.593 overall, 0.912 for interior reference cells and 0.212 for cells within 3 px of the field boundary, and for matched cells the error lay in the predicted phase level rather than in the predicted cell domain. On the 107 test fields whose in-line Gabor hologram is correctly paired with the reference (3055 reference cells), a single in-line training run reached a phase Pearson  $r$  of 0.8667 and a dry-mass MAPE of 0.2129; the tested classical in-line reconstruction recovered near-background specimen contrast ( $r = -0.2075$ ). On the off-axis test set, both measurement-aware integrated-phase terms increased the matched-cell dry-mass MAPE (+0.0141 per cell and +0.0145 per image, both resolved under a between-seed criterion), and their effect on the field-total error was not resolved. The forward-model consistency term did not resolve the phase scale under the implemented operator and did not constrain the off-axis propagation distance, whereas the in-line residual had its minimum at the recording distance. Displacing every reference boundary outward by one pixel, with no trained model involved, changed dry mass by 4.8% while Dice stayed at 0.974.

## 1 Introduction

Quantitative phase imaging measures the optical path length that a specimen adds to the illumination. For a cell this path length is proportional to the dry mass of its non-aqueous content [1–3], so a per-cell measurement in digital holographic microscopy (DHM) comes out of a chain of steps: the hologram is reconstructed into a phase map, each cell is delineated, and the phase is integrated inside each boundary. Dry mass is the product of a phase level and an integration domain. A phase map that is biased inside cells gives a biased mass through a correct boundary, and a correct phase map gives a biased mass through a boundary placed a pixel or two off. The scale of the second effect can be measured without any trained model: on the test fields used here, moving every reference boundary outward by one pixel changes dry mass by 4.8% while Dice stays at 0.974 (Sec. 5.6). Image-level statistics can also cancel the errors that a per-cell measurement is sensitive to: if one cell in a field is over-estimated by 20% and another is under-estimated by 20%, the field total is unbiased. Whether a reconstruction or a segmentation is adequate for quantitative work therefore has to be judged on the measurement itself.

The classical reconstruction chain has several stages, each with its own parameters, typically sideband filtering, phase unwrapping, aberration compensation and refocusing [4–6]. A recent review of learned methods in DHM organises the field along the same stages, with networks used for phase reconstruction and, separately, for analysis of the reconstructed phase [7]. Networks that take the recorded hologram as input can replace the reconstruction with a single forward pass [4, 8–10], or skip it for a downstream task such as detection, counting or classification [10–12]. Among the learned reconstruction studies we reviewed, the output is mostly scored against the classical phase with image criteria: SSIM, the fraction of pixels within a phase tolerance, or mean absolute error [4, 8, 10]. Two of them also compared cell measurements obtained from network and reference images, as distributions over the test cells [9, 13]. A roadmap on holographic QPI called for metrology studies of network-formed phase images before such images are accepted for quantitative imaging [14].

In most of the studies we examined, cells are segmented for per-cell measurement on phase that was reconstructed beforehand, either with classical rules [6, 15, 16] or with trained networks [17, 18]. Analysis directly on the hologram has also been reported. A convolutional network detected and counted red blood cells in raw off-axis holograms [11], a U-Net segmented cells on raw off-axis holograms with Dice and IoU of about 0.5 [12], and a network trained on simulated in-line holograms of microplankton predicts segmentation and dry-mass maps from the raw hologram [19]. Two systems cover more of the chain. In one, a single-shot Gabor hologram is mapped to quantitative phase and virtual fluorescence, per-cell measurements are extracted both with segmentation of the phase image and with membrane-fluorescence segmentation applied to the phase; measurements from generated and ground-truth images are compared as distributions, and the generated measurements are used for cell-type and drug-response classification [13]. In the other, an imaging flow cytometer chains a learned off-axis reconstruction with a cell detector and an aggregate analysis [10].

The components of a raw-hologram-to-measurement pipeline, and several partial combinations of them, therefore exist in prior work. In the studies reviewed here, we did not find paired per-cell measurement errors reported together with detection coverage. Nor did we identify a study that tests whether an explicit measurement term or a forward-model consistency term, added to a supervised objective, reduces the paired per-cell measurement error on experimental holographic data.

This study examines one network that takes a raw hologram and jointly predicts quantitative phase, a cell segmentation and, optionally, the transmitted amplitude. A fixed measurement chain computes per-cell quantities from these outputs, and no classical reconstruction enters the inference path. We keep three levels of evaluation separate: the network outputs (phase and amplitude error, segmentation overlap), cell detection and geometry (recall, precision, projected area, circularity), and the quantitative measurements (integrated phase and dry mass). The reference segmentation is derived from the reference phase by a fixed thresholding rule, so segmentation, detection and matched-cell results are relative to phase-derived labels.

The main contributions of this work are:

1. An evaluation of a joint raw-hologram-to-phase-and-mask network, built from a MobileNetV2 encoder [20] and U-Net-style decoders [21], by the paired per-cell error of matched cells, reported with detection recall, a coverage-adjusted error and field totals, and split into a domain factor and a phase factor (Secs. 4.5 and 5.2).
2. A comparison with one specified classical reconstruction pipeline through the same measurement chain, off-axis on 113 test fields and for both geometries on the 107 fields whose in-line hologram is correctly paired (Sec. 5.1).
3. Tests of measurement-aware integrated-phase objectives with three training seeds per configuration,

and of a forward-model consistency term, including its sensitivity to the phase scale and to the propagation distance in both geometries (Secs. 5.3 and 5.5).

4. A model-free analysis of how boundary displacement propagates into area and dry-mass error, and of the numerical error of the measurement chain on synthetic cells with analytic mass (Sec. 5.6).

## 2 Related work

### 2.1 Neural holographic phase reconstruction

Supervised networks have been trained to replace the numerical reconstruction of off-axis holograms. PhaseNet, a residual U-Net-style network, maps raw holograms of osteoblastic cells to phase [8]. HoloPhaseNet maps single-cell hologram crops to phase with a conditional GAN [9], and VY-Net maps raw holograms of living red blood cells to unwrapped, background-corrected phase [4]. OAH-Net builds Fourier-domain filtering and unwrapping into its layers and is the reconstruction stage of an imaging flow cytometer [10]. Other networks replace a single step of the classical chain, for example unwrapping and refocusing of reconstructed phase [22]. In the in-line geometry, where the twin image overlaps the object field [3], a diffusion-based network has been trained on unpaired single-shot Gabor holograms and off-axis phase images to recover quantitative phase [23], and a residual U-Net has been trained on procedurally generated in-line holograms of synthetic cells [24].

The phase target in these studies is a classical reconstruction or a simulation, and the reconstruction is scored with image criteria: SSIM against the target phase [4, 8, 22, 24], PSNR and SSIM [23], the fraction of pixels within 0.05 rad [4], or mean absolute error [10]. The VY-Net authors introduced their pixel-level indices because SSIM alone gave an incomplete assessment [4]. HoloPhaseNet also compared the distributions of projected area, circularity and dry mass computed from binarised network and numerical phase images over the test cells; paired per-cell errors are not reported [9].

### 2.2 Forward-model consistency in reconstruction

A second group of methods uses a model of hologram formation during training or reconstruction. Ongie *et al.* classify learned inverse solvers by whether and when they use the forward model. In their account, a known model used in training spares the network from learning the physics, and a model that depends on unknown calibration parameters is only partially known [25]. PhysenNet optimises an untrained network for each measurement so that angular-spectrum propagation of its phase reproduces the recorded in-line intensity at a fixed distance [26]. GedankenNet is trained on synthetic multi-height in-line holograms with a loss that compares the input holograms with holograms re-computed from the predicted field; its experimental distances come from a separate autofocus step [27]. HDPhysNet adds to a supervised phase prior a data-fidelity term that re-propagates the estimated object to the hologram plane, again with the distance obtained by autofocus [28]. In all three the term constrains agreement between the predicted field and the recorded in-line intensity at a distance that is known or estimated beforehand, and the reconstruction is evaluated with image metrics or phase error [26–28]. OAH-Net places the known off-axis reconstruction steps in the architecture instead of the loss [10]. The forward-model term studied here applies a hologram-consistency constraint to a supervised network trained on off-axis holograms and, in one configuration, makes the propagation distance a trainable parameter.

### 2.3 Cell segmentation, detection and dry-mass measurement in DHM

Classical pipelines segment reconstructed phase with thresholds, gradient criteria and watershed, and report per-cell area, optical volume or dry mass [6, 15, 16]. The generalist network Cellpose has also been applied to classically reconstructed phase [29]. Three of the learned segmentation methods we reviewed also work on reconstructed phase. DL-CSPF predicts a foreground mask and cell seed points with two networks and separates cells by marker-controlled watershed; its “physical framework” is this

decomposition, not an optical model [17]. HoloFluoNet predicts viability masks, a distance map for watershed separation and virtual nuclei from a single phase image [18], and zero-shot SAM has been used to segment red blood cells in DHM images [30]. Segmentation in these studies is scored with pixel-level or detection-level metrics [17, 18, 30], and the per-cell quantities are reported as time courses or distributions, without an error against a reference measurement of the same cell [15, 17, 18].

Analysis on the raw hologram, without reconstruction, has been reported since at least 2019. Trujillo and Garcia-Sucerquia trained a ten-layer convolutional network on sections of raw off-axis holograms and detected and counted red blood cells with a sliding window, validated against manual counts on reconstructed images [11]. A U-Net segmenting cells directly on raw off-axis holograms reached Dice and IoU of about 0.5 [12]. In RT-HAD, a detector applied to raw holograms was less accurate and slower than the same detector applied to reconstructed phase [10]. None of these raw-hologram analyses reports a per-cell phase measurement.

Per-cell dry mass follows from the integrated phase through the specific refraction increment  $\alpha$  [1, 2]. Reported values are  $0.190 \pm 0.003$  mL/g over a large protein set, with individual values between 0.173 and 0.215 [31], and  $0.18\text{--}0.21$   $\mu\text{m}^3/\text{pg}$  for biological media [32]. DHM studies have used 0.18 mL/g for haemoglobin [7], 0.2 mL/g [9, 15, 17] and 0.21 mL/g for microplankton [19].

## 2.4 Joint and end-to-end holographic pipelines

Three systems in the literature we reviewed come closest to a single path from hologram to cell measurement. The RU-Net of Bachimanchi *et al.* takes raw lens-less in-line holograms of microplankton and predicts, through two upsampling paths of one network, a segmentation map and maps of axial position and dry mass. It is trained only on simulated holograms and produces no phase image, and its dry-mass estimates are validated per species against volume-to-carbon estimates and elemental analysis [19]. The Gabor dual-mode system of Park *et al.* reconstructs quantitative phase and virtual fluorescence channels from an experimental single-shot Gabor hologram with a diffusion generator trained on unpaired data; its main models take  $256 \times 256$  inputs. Its phase targets come from off-axis holography and its fluorescence targets from an epifluorescence path of the same instrument. Optical path difference, dry mass, projected area and circularity are extracted from both the reconstructed and the ground-truth images with two segmentation strategies, segmentation of the phase image and membrane-fluorescence segmentation applied to the phase; measurements from generated and ground-truth images are compared as distributions, and the generated measurements are used for cell-type and drug-response classification. The model has 96.97 M parameters and runs at 89.47 ms per image [13]. RT-HAD processes raw off-axis holograms with a chain of separately trained models (learned reconstruction, bounding-box detection, aggregate analysis) at under 10 ms per frame. It validates cell-type ratios and aggregate counts against a haematology analyser and reports no per-cell area or dry mass [10]. Earlier, a roadmap on holographic QPI had proposed networks that map intensity acquisitions directly to images encoding cell position and dry mass [14].

These systems differ from one another, and from the network studied here, in what enters the network, what it outputs and how the measurement is checked. RU-Net starts from the raw hologram but outputs no phase map. RT-HAD reconstructs phase but detects cells as bounding boxes and reports no per-cell phase quantity. The Gabor system outputs phase and segments it either directly or with the help of membrane fluorescence. In all three the measurement is compared as a distribution, per species or per population. The cell domains are supervised by simulation in RU-Net, by phase-image or fluorescence segmentation in the Gabor system and by manually drawn boxes in RT-HAD.

## 2.5 Measurement-aware training objectives

In the studies we reviewed, a per-cell measurement enters training only when it is itself the regression target. This is the case for the dry-mass outputs of RU-Net and of WAC-Net, the second network in the same study, both learned from simulated ground truth [19]. Elsewhere the losses are pixel-wise, adversarial or contrastive, and per-cell quantities are computed after inference [9, 13, 17, 18]. Pixel-wise mean squared error does not fully reflect differences in reconstruction quality, and the alternative losses usually proposed for learned inversion target perceptual quality [33]. Ongie *et al.* note that realistic-looking artefacts are a particular concern when reconstructions feed downstream tasks [25]. The integrated-phase terms tested here keep the pixel-wise supervision and add a penalty on the phase integrated over reference cell domains or over the image.

## 2.6 Position of this study

Among the studies reviewed here, raw-hologram-to-phase reconstruction [4, 8, 9, 23], detection and segmentation on raw holograms [10–12], dry mass predicted from raw holograms [19] and phase-derived per-cell measurement from a learned reconstruction [9, 13] have each been reported. The present study combines these steps in one jointly trained network with 9.6 M parameters. The network is trained on experimental off-axis and in-line holograms, with a classical reconstruction as the phase target and segmentation labels derived from that phase. We evaluate the whole chain by the paired per-cell error of matched cells, reported together with detection coverage, and test measurement-aware and forward-model terms against that error.

## 3 Method

### 3.1 Overall framework

Figure 1 summarises the pipeline. A raw off-axis or in-line Gabor hologram, cropped to the  $900 \times 900$  phase grid (Sec. 4.1), enters a network with one MobileNetV2 encoder and two U-Net decoders (Fig. 1a,b). The network predicts quantitative phase, a two-class segmentation and, in the +Amplitude and +Fwd configurations only, a transmitted-amplitude map (Fig. 1c). Watershed instances of the predicted mask define the domains over which a fixed measurement chain computes projected area, circularity, integrated phase and dry mass from the predicted phase (Fig. 1d). The training configurations add terms to a baseline objective, with reference masks derived from the reference phase (Fig. 1e). The per-cell quantities are compared with the same quantities computed from the reference data and from a classical pipeline, and model-free analyses bound the error of the measurement chain itself (Fig. 1f).

### 3.2 Hologram formation and propagation model

A thin specimen illuminated by a plane wave transmits  $U = A e^{i\varphi}$ , with  $A$  the transmitted amplitude and  $\varphi$  the phase. Propagation over a distance  $z$  is modelled with the angular-spectrum operator

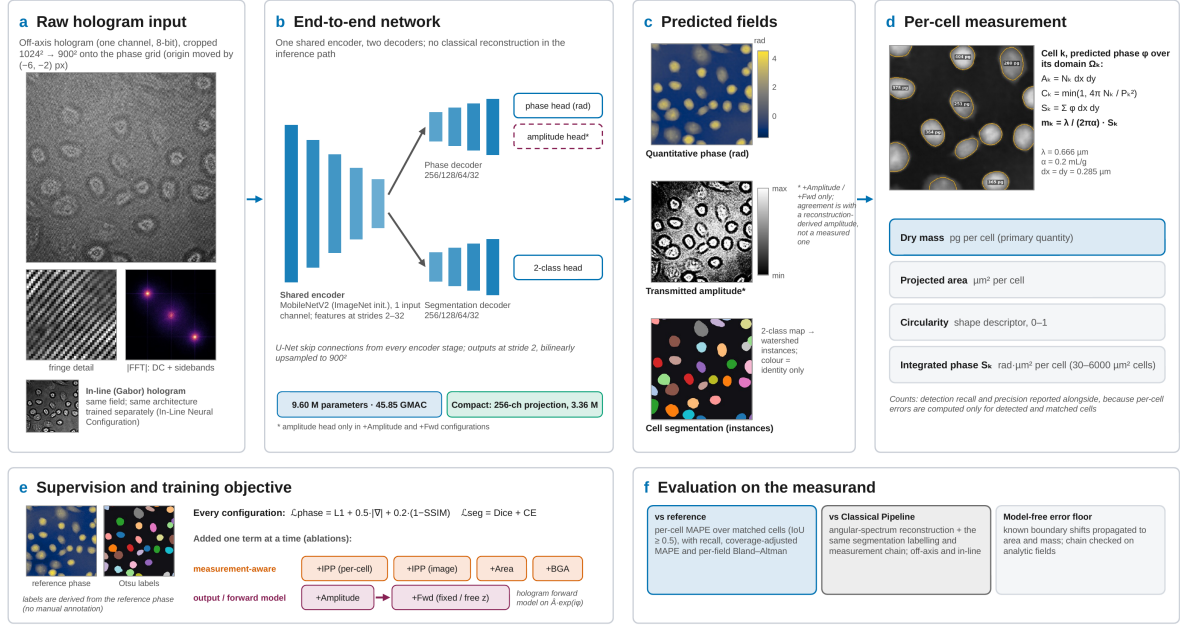
$$\mathcal{P}_z\{U\} = \mathcal{F}^{-1}\left\{\mathcal{F}\{U\} e^{i2\pi z \sqrt{\lambda^{-2} - f_x^2 - f_y^2}}\right\}, \quad (1)$$

with evanescent components attenuated as  $e^{-2\pi|z|\sqrt{f_x^2 + f_y^2 - \lambda^{-2}}}$ . In the in-line (Gabor) geometry the sensor records

$$I_{\text{in}} = |\mathcal{P}_z\{U\}|^2, \quad (2)$$

which tends to  $A^2$  as  $z \rightarrow 0$  and so carries phase only through propagation. In the off-axis geometry a tilted reference  $R$  is added,

$$I_{\text{off}} = |R|^2 + |\mathcal{P}_z\{U\}|^2 + R^* \mathcal{P}_z\{U\} + R \mathcal{P}_z\{U\}^*, \quad (3)$$



**Figure 1:** Overview of the evaluated pipeline: (a) raw hologram input, cropped to the phase grid; (b) network; (c) predicted outputs; (d) per-cell measurement; (e) training configurations; (f) evaluation. Panels a, c and d show test field NCI\_08; phase, segmentation and measurement are from the seed-42 End-to-End Neural Baseline, and the amplitude map from +Amplitude.

and the object field appears in a carrier-modulated cross-term that can be isolated in the Fourier plane at any  $z$  [34]. The classical pipeline (Sec. 4.4) and the forward-model term (Sec. 3.5.4) use these relations. The network itself receives only the recorded intensity.

### 3.3 Network architecture

A MobileNetV2 encoder [20], initialised from ImageNet with its first convolution rebuilt for one input channel by averaging the RGB filters, produces feature maps at strides 2–32. Two U-Net decoders [21] with widths 256/128/64/32 consume them: a phase decoder that carries the phase head and, when enabled, the amplitude head, and a segmentation decoder with a two-class head. The phase head has no output activation and emits radians. The amplitude head outputs  $1 + 0.3 \tanh(\cdot)$  and is zero-initialised, so it starts at unit transmittance. The input is  $z$ -scored per field and reflection-padded to a multiple of 32. The decoders end at stride 2, so the phase and segmentation maps are produced at half resolution and bilinearly upsampled to  $900 \times 900$ . The standard model has 9 598 099 parameters (45.85 GMAC at  $900 \times 900$ ). A compact variant inserts a shared  $1 \times 1$  projection from 1280 to 256 channels before both decoders, which reduces the count to 3 360 403.

### 3.4 Per-cell measurement domains and quantitative measurements

Cell domains are the instances of a binary mask separated by a distance-transform watershed (minimum peak distance 15 px) and kept if their area lies in 30–6000  $\mu\text{m}^2$ . The same instance labelling is applied to predicted and reference masks. For each instance  $\Omega_k$  with  $N_k$  pixels and Crofton perimeter  $P_k$  (in pixels),

$$A_k = N_k dx dy, \quad (4)$$

$$C_k = \min(1, 4\pi N_k / P_k^2), \quad (5)$$

$$S_k = \sum_{p \in \Omega_k} \varphi_p dx dy, \quad (6)$$

$$m_k = \frac{\lambda}{2\pi\alpha} S_k, \quad (7)$$

with  $dx = dy = 0.284871 \mu\text{m}$ ,  $\lambda = 0.666 \mu\text{m}$  and  $\alpha = 0.2 \text{ mL/g}$ .  $S_k$  is the integrated phase in  $\text{rad } \mu\text{m}^2$ , and Eq. (7) is the dry-mass relation of [1]. Circularity is evaluated on the pixel grid, which is exact here because the sampling is isotropic. Predicted measurements use the predicted phase inside predicted instances, and reference measurements use the reference phase inside reference instances, with the same code. Predicted and reference instances are paired one-to-one by greedy descending IoU at  $\text{IoU} \geq 0.5$ .

### 3.5 Training objectives

The objective is

$$\mathcal{L} = \mathcal{L}_{\text{phase}} + \mathcal{L}_{\text{seg}} + w_{\text{IPP}}^{\text{cell}} \mathcal{L}_{\text{IPP}}^{\text{cell}} + w_{\text{IPP}}^{\text{img}} \mathcal{L}_{\text{IPP}}^{\text{img}} + w_{\text{area}} \mathcal{L}_{\text{area}} + w_{\text{BGA}} \mathcal{L}_{\text{BGA}} + w_{\text{amp}} \mathcal{L}_{\text{amp}} + w_{\text{fwd}} \mathcal{L}_{\text{fwd}}, \quad (8)$$

and every weight other than those of  $\mathcal{L}_{\text{phase}}$  and  $\mathcal{L}_{\text{seg}}$  is zero in the baseline. The terms fall into two groups. The measurement-aware objectives ( $\mathcal{L}_{\text{IPP}}^{\text{cell}}$ ,  $\mathcal{L}_{\text{IPP}}^{\text{img}}$ ,  $\mathcal{L}_{\text{area}}$ ,  $\mathcal{L}_{\text{BGA}}$ ) encode the structure of the measurement of Eq. (7) and use no model of image formation. The forward-model consistency term  $\mathcal{L}_{\text{fwd}}$  is the only physics-based term: it compares the prediction with the recorded hologram through Eqs. (1) to (3).

#### 3.5.1 Supervised reconstruction and segmentation losses

The supervised terms are

$$\mathcal{L}_{\text{phase}} = \|\hat{\phi} - \phi\|_1 + 0.5 \|\nabla \hat{\phi} - \nabla \phi\|_1 + 0.2 (1 - \text{SSIM}(\hat{\phi}, \phi)), \quad (9)$$

$$\mathcal{L}_{\text{seg}} = \text{Dice} + \text{CE}, \quad \text{class weights } (0.5, 1.0), \quad (10)$$

with pixel-mean norms, forward-difference gradient magnitude, and SSIM over an 11-pixel Gaussian window ( $\sigma = 1.5$ , data range 8 rad).

#### 3.5.2 Measurement-aware objectives

The measurement-aware terms couple the predicted phase to the predicted foreground probability  $f = 1 - p_{\text{background}}$  from the segmentation head. With  $M$  the reference mask,  $\Omega_k$  the reference instances of a training crop, and  $\varepsilon = 10^{-4}$ , the per-cell integrated-phase-preservation (IPP) term is

$$\mathcal{L}_{\text{IPP}}^{\text{cell}} = \frac{1}{K} \sum_{k=1}^K \frac{|\sum_{p \in \Omega_k} f_p \hat{\phi}_p - \sum_{p \in \Omega_k} M_p \phi_p|}{|\sum_{p \in \Omega_k} M_p \phi_p| + \varepsilon}, \quad (11)$$

over the  $K$  instances whose reference integral is at least 50 rad px, with each relative error capped at 10. The domains  $\Omega_k$  are the watershed instances of the reference mask, computed on the full field before cropping, so an instance cut by a training crop is integrated over its visible part. Because the domains are fixed reference instances, the term cannot be reduced by moving them, and predicted foreground outside every reference instance does not enter it. The image-level term replaces the per-cell sums with sums over the whole crop,

$$\mathcal{L}_{\text{IPP}}^{\text{img}} = \frac{|\sum_p f_p \hat{\phi}_p - \sum_p M_p \phi_p|}{|\sum_p M_p \phi_p| + \varepsilon}. \quad (12)$$

On a field with one cell over-measured by 20% and another under-measured by 20%, Eq. (12) is 0 and Eq. (11) is 0.2 (a unit test in the repository). Since the quantity in Eq. (11) depends on both the predicted foreground probability and the predicted phase inside  $\Omega_k$ , the term can, as a formal property, be satisfied by a foreground that recedes inside the reference domain together with a raised phase. The per-cell area

term constrains the domain directly,

$$\mathcal{L}_{\text{area}} = \frac{1}{K'} \sum_{k=1}^{K'} \frac{|\sum_{p \in \Omega_k} f_p - \sum_{p \in \Omega_k} M_p|}{\sum_{p \in \Omega_k} M_p + \varepsilon}, \quad (13)$$

over the  $K'$  instances of at least 370 px ( $30 \mu\text{m}^2$ ). It cannot penalise foreground outside reference instances. The boundary–gradient alignment term is

$$\mathcal{L}_{\text{BGA}} = \left\| \frac{|\nabla f|}{\max |\nabla f| + \varepsilon'} - \frac{|\nabla \hat{\phi}|}{\max |\nabla \hat{\phi}| + \varepsilon'} \right\|_1, \quad \varepsilon' = 10^{-8}. \quad (14)$$

These four terms are averaged over the crops of a batch that contain at least 370 reference-foreground pixels.

Measured before training over 30 batches of  $512 \times 512$  crops, the gradient of  $\mathcal{L}_{\text{IPP}}^{\text{cell}}$  into the segmentation decoder at unit weight has a median magnitude of 0.373 (range 0.243–0.655) times that of  $\mathcal{L}_{\text{seg}}$  on the same parameters. The central configuration uses  $w_{\text{IPP}}^{\text{cell}} = 1.0$ , and a sweep covers 0.1, 0.3, 1.0 and 3.0.  $\mathcal{L}_{\text{IPP}}^{\text{img}}$  is used at the same coefficient,  $w_{\text{IPP}}^{\text{img}} = 1.0$ ; its gradient magnitude was not matched to that of  $\mathcal{L}_{\text{IPP}}^{\text{cell}}$ .

### 3.5.3 Amplitude objective

The amplitude term  $\mathcal{L}_{\text{amp}}$  is the pixel-mean  $\ell_1$  distance between the predicted amplitude and the amplitude reference of Sec. 4.2.

### 3.5.4 Forward-model consistency

$\mathcal{L}_{\text{fwd}}$  is the only term that compares the prediction with the recorded hologram. The predicted field  $\hat{A} e^{i(\hat{\phi} + \Psi)}$  is reflection-padded, propagated by Eq. (1), and expanded into the intensity components of Eq. (2) or Eq. (3).  $\Psi$  is an order-5 polynomial aberration surface. It is added back because the supplied phase had it removed while the recorded hologram did not. A single surface, the per-coefficient median of per-field fits over the training split, is used for every field, since per-field surfaces are fitted against the reference phase and cannot be obtained at inference. The unknown radiometric constants are removed by a per-image, ridge-regularised least-squares fit of the recorded intensity  $H$  onto the components  $B_j$ :  $\{1, |\mathcal{P}_z|^2\}$  in-line, and off-axis also the real and imaginary parts of  $R^* \mathcal{P}_z$  and  $R \mathcal{P}_z$ , with  $R$  a plane wave at the carrier frequency estimated from  $H$ . Carrying both conjugate cross-terms and both quadratures makes the fit independent of which sideband is selected and of the global piston phase  $e^{i2\pi z/\lambda}$ . With  $\hat{H} = \sum_j c_j B_j$  the fitted intensity,

$$\mathcal{L}_{\text{fwd}} = \frac{\sum_p (\hat{H}_p - H_p)^2}{\sum_p (H_p - \bar{H})^2}, \quad (15)$$

evaluated after discarding a border of 79 px (the diffraction spread at the configured  $z$ ).  $\mathcal{L}_{\text{fwd}}$  is the fraction of the hologram’s variance that the synthesised hologram leaves unexplained. The propagation distance is fixed at the supplied  $z = 33.77 \mu\text{m}$  or, in one configuration, made a trainable parameter with its own learning rate ( $100\times$  the base rate, no weight decay). The pad and border stay fixed at the configured distance so that the comparison domain does not change with  $z$ . Everything  $\mathcal{L}_{\text{fwd}}$  uses is available at inference: the carrier frequency and radiometric coefficients come from the recorded hologram, and the global aberration surface from a one-time calibration on training fields. A margin of 0.01 in residual units is the configured tolerance below which a difference in  $\mathcal{L}_{\text{fwd}}$  is treated as noise (Sec. 5.5).

## 4 Experiments

### 4.1 Dataset and imaging configuration

The dataset contains 800 fields of view of three human cancer cell lines (NCI-H1299, SNU-475 and T-24; NCI, SNU and T24 below), each under a control condition or one of four drugs: blebbistatin (myosin II inhibition), FCCP (mitochondrial uncoupling), staurosporine (apoptosis) and rotenone (complex I inhibition). Table 1 gives the number of fields per cell line and condition. The drug condition is a property of the dish, so it labels the field, not the individual cell; it is not a network input or target and is used only to stratify the split. Each field has an off-axis hologram and an in-line Gabor hologram ( $1024 \times 1024$ , 8-bit), a  $900 \times 900$  quantitative phase map stored as float32 in radians, and a membrane-stain fluorescence image from a separate camera. The illumination wavelength is  $0.666 \mu\text{m}$ , the pixel pitch at the sample  $0.284871 \mu\text{m}$ , and the supplied propagation distance  $33.77 \mu\text{m}$ . The phase maps were supplied with the dataset as reconstructions that are already aberration-corrected, background-subtracted and unwrapped; the off-axis reconstruction principle is that of [34], and we do not know the exact processing chain that produced them. We use them as the phase reference. Fields are split 560/127/113 (train/validation/test), stratified by cell line  $\times$  condition, and the same split is used for every off-axis configuration and seed. The 113 test fields contain 3186 reference cells. Acquisition, calibration and training settings are listed in Table 2.

**Table 1:** Dataset composition: number of fields of view per cell line and drug condition.

| Cell line | Ctrl | Bleb | FCCP | Stau | Rot | Total      |
|-----------|------|------|------|------|-----|------------|
| NCI       | 50   | 50   | 50   | 50   | 50  | 250        |
| SNU       | 100  | 50   | 50   | 50   | 50  | 300        |
| T24       | 50   | 50   | 50   | 50   | 50  | 250        |
| Total     | 200  | 150  | 150  | 150  | 150 | <b>800</b> |

Ctrl, control; Bleb, blebbistatin; Stau, staurosporine; Rot, rotenone.

The hologram is cropped from 1024 to 900 px without resampling, with the crop origin moved by  $(-6, -2)$  px in  $(y, x)$  from the exactly centred crop. The origin was determined by registering the classical off-axis reconstruction (Sec. 4.4) of each field to its reference phase. On the 127 validation fields, the centred crop left a median offset of  $+5.4$  px in  $y$  and  $+1.1$  px in  $x$ . The rounded estimate and its eight neighbouring whole-pixel origins were then evaluated on the same validation fields, and  $(-6, -2)$  gave median offsets of 0.0 and 0.0 px. On the 113 test fields the median residual offset at this origin is  $-0.1$  px in  $y$  (5–95% range  $-1.88$  to  $+1.64$  px) and 0.0 px in  $x$  ( $-3.06$  to  $+2.12$  px). The same crop is used for the network input, the forward model, the aberration fit, the amplitude reference and both classical pipelines.

The in-line holograms of fields SNU\_01 to SNU\_50 do not show the same field of view as their off-axis holograms and reference phase. This was established by registering each in-line frame to its off-axis hologram after band-pass filtering (a Gaussian of  $\sigma = 5$  px minus one of  $\sigma = 50$  px, with a Hann window), against a control in which every in-line frame was paired with a different field’s off-axis hologram by a random derangement. Matched pairs were shifted by at most 1 px on 734 of the 800 fields and control pairs on 3 of 800, and the correlation after registration separated matched from deranged pairs with an AUC of 0.975 (0.973 by registration error). The 750 correctly paired fields all registered within 4.7 px, 734 of them within 1 px, with a correlation after registration of 0.47–1.00, above the highest control value (0.25). The 50 fields SNU\_01 to SNU\_50 gave correlations of  $-0.13$  to  $0.18$ , at the level of the unmatched control. A low-pass filter alone ( $\sigma = 5$  px) did not separate matched from unmatched pairs (AUC 0.506), so the band-pass variant was used. The off-axis holograms of these 50 fields do match the reference phase. They were therefore excluded from the in-line modality only, before

training, which leaves an in-line split of 521/122/107 fields; every off-axis configuration keeps all 800 fields. Comparisons across geometries use the 107 test fields common to both (3055 reference cells).

**Table 2:** Acquisition, calibration, data split and training settings.

|                                         |                                                                      |
|-----------------------------------------|----------------------------------------------------------------------|
| <b>Acquisition and calibration</b>      |                                                                      |
| Hologram                                | 1024 × 1024, 8-bit TIFF                                              |
| Quantitative phase (reference)          | 900 × 900, float32, radians                                          |
| Hologram-to-phase mapping               | crop 1024 → 900, origin (−6, −2) px from the centred crop            |
| Illumination wavelength $\lambda$       | 0.666 $\mu\text{m}$                                                  |
| Pixel pitch $dx = dy$                   | 0.284871 $\mu\text{m}$                                               |
| Refraction increment $\alpha$           | 0.2 mL/g                                                             |
| Propagation distance $z$                | 33.77 $\mu\text{m}$ (supplied)                                       |
| Aberration surface (forward model)      | order-5 polynomial, median over training fields                      |
| <b>Splits</b>                           |                                                                      |
| Off-axis fields (train / val / test)    | 560 / 127 / 113                                                      |
| In-line fields (train / val / test)     | 521 / 122 / 107 (SNU_01–SNU_50 excluded)                             |
| Stratification                          | cell line × drug condition                                           |
| Reference cells in test                 | 3186 off-axis; 3055 in-line                                          |
| <b>Network</b>                          |                                                                      |
| Encoder                                 | MobileNetV2, ImageNet-initialised                                    |
| Decoders                                | two U-Net decoders, 256/128/64/32                                    |
| Heads                                   | phase, amplitude (optional), segmentation (2 classes)                |
| Network input                           | raw hologram, z-scored per field                                     |
| Output grid                             | stride 2, bilinearly upsampled                                       |
| Parameters, standard                    | 9 598 099 (45.85 GMAC)                                               |
| Parameters, with amplitude              | 9 607 412 (47.87 GMAC)                                               |
| Parameters, compact                     | 3 360 403 (24.03 GMAC)                                               |
| <b>Optimisation</b>                     |                                                                      |
| Epochs                                  | 60                                                                   |
| Optimiser                               | AdamW, $\beta = (0.9, 0.999)$                                        |
| Learning rate                           | $3 \times 10^{-4}$ , 2 warm-up epochs, cosine to $10^{-6}$           |
| Learning rate for $z$ (+Fwd, free $z$ ) | 100× base                                                            |
| Weight decay                            | $1 \times 10^{-4}$                                                   |
| Batch size / train crop                 | 4 / 512 × 512                                                        |
| Precision                               | mixed, FP16                                                          |
| Gradient clipping                       | 1.0 (global norm)                                                    |
| Augmentation                            | flips, 90° rotations                                                 |
| Checkpoint criterion                    | best validation composite (Sec. 4.8), every epoch, no early stopping |
| Seeds                                   | 42 (primary), 1337, 2024                                             |

## 4.2 Reference phase, segmentation labels and amplitude reference

No manual annotation is available, so the reference masks are derived from the reference phase: Gaussian smoothing ( $\sigma = 4$  px), a per-field Otsu threshold, binary closing (2 px), hole filling, removal of objects touching the array border, and the area filter of Sec. 3.4. Instances are separated with the watershed of Sec. 3.4. These are phase-derived, silver-standard labels: the segmentation target is a deterministic function of the phase target. Two further properties of the construction matter for the results. The closing step leaves an empty two-pixel rim at the array border, so the border-removal step, as configured, removes no objects, and objects truncated by the field edge and elongated regions along the field boundary remain in the reference (Fig. 3; quantified in Sec. 5.2). The Otsu level also varies across fields ( $0.406 \pm 0.151$  rad over 800 fields) and differs between drug conditions (one-way ANOVA  $F = 15.26$ ,  $p = 4.9 \times 10^{-12}$ ).

As a check that does not depend on the phase, the membrane channel was registered to the phase grid (scale 1.343769, equal to the ratio of the phase pixel pitch, 0.284871  $\mu\text{m}$ , to the second pitch value stored in the phase-file header, 0.211994  $\mu\text{m}$ , which the registration search also preferred on all 13 fields tested;

fitted offset 314/280 px in  $y/x$  on the phase grid), Gaussian-smoothed ( $\sigma = 2$  px), thresholded at the 75th percentile of the illuminated region, opened with a  $3 \times 3$  structuring element and hole-filled. The stain marks cell perimeters, so these masks are provisional. They are used only in Sec. 5.2.

No measured amplitude exists for these data. For the configurations with an amplitude output, the target is the modulus of a classical off-axis reconstruction (first-order sideband isolation) of the cropped hologram, normalised to a background median of 1 and clipped at 1.5. It is a reconstruction-derived reference, not ground truth.

### 4.3 Training and experimental configurations

Table 3 lists the configurations. Each differs from its comparator by one term, one output or one capacity choice. The End-to-End Neural Baseline uses  $\mathcal{L}_{\text{phase}} + \mathcal{L}_{\text{seg}}$  on off-axis holograms, and the In-Line Neural Configuration applies the same objective to the in-line holograms of the 750 correctly paired fields; the two geometries are trained separately and share no weights. +IPP (per-cell) and +IPP (image) each add one integrated-phase term to the baseline at the same coefficient,  $w = 1.0$ , so their comparison is an equal-coefficient comparison in which the gradient magnitudes were not matched. +Area and +BGA each add one term to +IPP (per-cell). +Amplitude adds the amplitude head and  $\mathcal{L}_{\text{amp}}$  ( $w = 0.1$ ) to +IPP (per-cell). +Fwd (fixed  $z$ ) adds  $\mathcal{L}_{\text{fwd}}$  ( $w = 0.02$ ) to +Amplitude, because the term uses the predicted complex field, and +Fwd (free  $z$ ) makes  $z$  trainable. The Compact Baseline and Compact +IPP are the baseline and +IPP (per-cell) with the compact decoder. All configurations are off-axis except the In-Line Neural Configuration. The  $w = 1.0$  member of the weight sweep is +IPP (per-cell) itself. Training settings are in Table 2.

**Table 3:** Experimental configurations. Each differs from its comparator by one change; weights not listed are zero. Training runs (seeds): number of trained models and their seeds.

| Configuration                                         | Change relative to comparator                                                          | Comparator                 | Training runs (seeds) |
|-------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------|-----------------------|
| <i>Main pipeline and hologram mode</i>                |                                                                                        |                            |                       |
| End-to-End Neural Baseline                            | off-axis; reconstruction + segmentation losses only                                    | –                          | 3 (42, 1337, 2024)    |
| In-Line Neural Configuration                          | the baseline’s objective on in-line Gabor holograms; SNU_01–SNU_50 excluded            | End-to-End Neural Baseline | 1 (42)                |
| <i>Measurement-aware objectives</i>                   |                                                                                        |                            |                       |
| +IPP (per-cell)                                       | $+\mathcal{L}_{\text{IPP}}^{\text{cell}}, w = 1.0$                                     | End-to-End Neural Baseline | 3 (42, 1337, 2024)    |
| +IPP (image)                                          | $+\mathcal{L}_{\text{IPP}}^{\text{img}}, w = 1.0$                                      | End-to-End Neural Baseline | 3 (42, 1337, 2024)    |
| +Area                                                 | $+\mathcal{L}_{\text{Area}}, \text{per-cell}, w = 1.0$                                 | +IPP (per-cell)            | 1 (42)                |
| +BGA                                                  | $+\mathcal{L}_{\text{BGA}}, w = 0.05$                                                  | +IPP (per-cell)            | 1 (42)                |
| +IPP (per-cell),<br>$w = 0.1/0.3/3.0$                 | $\mathcal{L}_{\text{IPP}}^{\text{cell}}$ weight 0.1, 0.3, 3.0                          | End-to-End Neural Baseline | 1 (42)                |
| <i>Amplitude output and forward-model consistency</i> |                                                                                        |                            |                       |
| +Amplitude                                            | $+\text{amplitude head}, +\mathcal{L}_{\text{amp}}, w = 0.1$                           | +IPP (per-cell)            | 1 (42)                |
| +Fwd (fixed $z$ )                                     | $+\mathcal{L}_{\text{fwd}}, w = 0.02, z$ fixed                                         | +Amplitude                 | 1 (42)                |
| +Fwd (free $z$ )                                      | $z$ becomes a trainable parameter                                                      | +Fwd (fixed $z$ )          | 1 (42)                |
| <i>Model capacity</i>                                 |                                                                                        |                            |                       |
| Compact Baseline                                      | shared $1 \times 1$ projection to 256 channels before the decoders (3.36 M parameters) | End-to-End Neural Baseline | 1 (42)                |
| Compact +IPP                                          | compact decoder, with $\mathcal{L}_{\text{IPP}}^{\text{cell}}$                         | +IPP (per-cell)            | 1 (42)                |

### 4.4 Classical comparison pipeline

The comparison is with one classical configuration per geometry, scored by the same instance labelling and measurement code. Off-axis, it is first-order sideband isolation (radius 130 px, DC exclusion 60 px), conjugate-sideband selection by phase skewness, and angular-spectrum back-propagation to  $33.77 \mu\text{m}$ .

In-line, it is back-propagation of the recorded amplitude ( $\sqrt{I}$ ) to  $33.77\mu\text{m}$  followed by 20 Gerchberg–Saxton iterations with the sample-plane modulus constrained to at most 1. Both are followed by phase unwrapping, subtraction of an order-3 polynomial fitted to background pixels of a preliminary mask, and the reference-mask rule of Sec. 4.2 applied to the resulting phase. The classical segmentation is thus the labelling rule itself applied to a classical reconstruction. Both pipelines use the crop of Sec. 4.1.

#### 4.5 Evaluation protocol

All metrics are computed at full  $900 \times 900$  resolution, on the 113 off-axis test fields unless stated, and the cross-geometry comparison on the 107 common test fields.

*Reconstruction.* Phase MAE over the field and inside reference cells, Pearson  $r$  and SSIM between predicted and reference phase describe the phase map, an intermediate output.

*Segmentation and detection.* Dice, AJI and boundary F1 at 2 px tolerance describe the mask, also an intermediate output. Detection recall and precision are computed from the IoU matching of Sec. 3.4.

*Per-cell measurement.* Mean absolute percentage error (MAPE) and per-cell Pearson  $r$  of projected area, circularity and dry mass are computed over matched cells only, so they describe how well detected cells are measured. Because  $\lambda/(2\pi\alpha)$  multiplies predicted and reference  $S_k$  alike, the relative error of the integrated phase of a cell equals its relative dry-mass error, and every relative error, correlation and relative bias of dry mass reported here is independent of  $\lambda$  and  $\alpha$ . We report it once, as dry-mass MAPE.

*Coverage and field totals.* Matched-cell errors are reported beside recall and a coverage-adjusted error that charges each missed reference cell as a 100% error,

$$\text{MAPE}_{\text{cov}} = r_{\text{det}} \cdot \text{MAPE} + (1 - r_{\text{det}}), \quad (16)$$

and beside field totals, which sum all predicted and all reference cells on a field without pairing; their error is reported as a field-total MAPE and a field-total bias. Method agreement is reported as a relative Bland–Altman analysis [35] on per-field medians of matched cells.

*Mass-error decomposition.* With  $S(\Omega, \varphi) = \sum_{p \in \Omega} \varphi_p dx dy$ , the ratio of predicted to reference dry mass factorises exactly,

$$\frac{\hat{m}}{m} = \underbrace{\frac{S(\hat{\Omega}, \varphi)}{S(\Omega, \varphi)}}_{\text{domain factor}} \times \underbrace{\frac{S(\hat{\Omega}, \hat{\varphi})}{S(\hat{\Omega}, \varphi)}}_{\text{phase factor}}, \quad (17)$$

where  $\hat{\Omega}$  and  $\Omega$  are the predicted and reference domains. The domain factor holds the phase at the reference and changes only the domain; the phase factor holds the domain at the prediction and changes only the phase. At cell level,  $\Omega$  is a reference instance and  $\hat{\Omega}$  the predicted instance matched to it, and cells are split into edge cells (a reference pixel within 3 px of the field boundary) and interior cells. At field level,  $\hat{\Omega}$  and  $\Omega$  are the whole predicted and reference foreground, so undetected and spurious cells enter the domain factor. Each factor is summarised by its geometric mean, which composes exactly, and by the median absolute log ratio as a sign-free measure of spread. The decomposition describes how the error is structured; it does not identify what causes it.

#### 4.6 Computational benchmarking protocol

Inference cost was measured at  $900 \times 900$  and batch 1 on one NVIDIA RTX A5000, in PyTorch 2.6 and in ONNX Runtime (opset 17, CUDA execution provider); the FP16 models are half-precision exports of weights and input. Each measurement is 50 warm-up passes followed by 500 timed passes. The baseline was benchmarked once per trained seed, every other configuration once. The  $p_{99}/p_{50}$  latency ratio is recorded as a check on timing stability, with a limit of 1.25. Weight size and peak allocated memory are reported for PyTorch only, because ONNX Runtime does not expose comparable allocator accounting;

the peak allocation does not separate this process’s CUDA context from other jobs on the device.

#### 4.7 Model-free sensitivity analyses

Boundary sensitivity is measured on the off-axis test fields by dilating or eroding each reference instance by 1–5 pixels and recomputing area and dry mass from the reference phase. The numerical error of the measurement chain is measured on synthetic fields of spherical-cap cells,  $\varphi(d) = \varphi_0 \sqrt{1 - (d/r)^2}$ , whose area  $\pi r^2$  and integrated phase  $\frac{2}{3}\pi r^2 \varphi_0$  are exact. Three independently generated sets are used.

#### 4.8 Statistical analysis

The baseline, +IPP (per-cell) and +IPP (image) were trained at seeds 42, 1337 and 2024, and every other configuration once, at seed 42. Values reported for the replicated configurations are three-run means with the between-seed sample SD. A difference between two configurations is called *resolved* if it exceeds twice the pooled between-seed standard deviation,  $2\sqrt{(s_1^2 + s_2^2)/2}$ , of the same metric. This is a separation criterion defined before the comparisons were made; no hypothesis test is performed, and a comparison in which either configuration has fewer than three runs is reported as not resolvable. Within-run intervals are percentile bootstrap intervals over fields (2000 resamples). Evaluation is deterministic in data and weights but not bit-exact, because convolution algorithms are selected at run time. Checkpoints were selected on the validation split by  $0.35 \text{ Dice} + 0.25 r_\varphi + 0.25 (1 - \text{MAPE}_{\text{mass}}) + 0.15 F1_{\text{det}}$ , so the validation dry-mass error enters model selection but not training. Location stratification (edge and interior) and the mass-error decomposition were computed after training, from the stored models and reference masks.

### 5 Results

#### 5.1 Reconstruction and segmentation from raw holograms

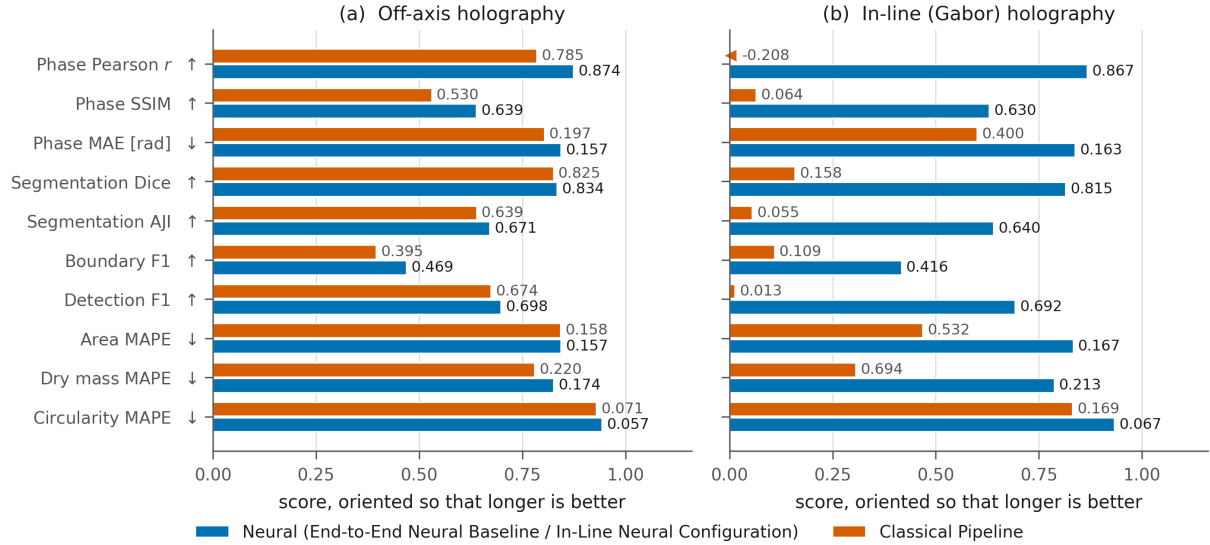
On the 113 off-axis test fields, the End-to-End Neural Baseline has, compared with the tested classical pipeline, lower phase error (MAE  $0.1968 \rightarrow 0.1568$  rad; inside reference cells  $0.3112 \rightarrow 0.2689$  rad), higher phase correlation ( $0.7851 \rightarrow 0.8741$ ) and higher SSIM ( $0.5299 \rightarrow 0.6386$ ) (Table 4, Fig. 2a). The segmentation scores are closer: Dice  $0.8253 \rightarrow 0.8337$ , AJI  $0.6387 \rightarrow 0.6710$  and boundary F1  $0.3954 \rightarrow 0.4687$ , a relative increase of 19%. Recall is nearly the same ( $0.5869$  and  $0.5931$ ) and precision is higher for the network ( $0.7910 \rightarrow 0.8488$ ). At the measurement level, projected-area MAPE is essentially equal ( $0.1578$  and  $0.1568$ ) while the matched-cell dry-mass MAPE is lower for the network ( $0.2203 \rightarrow 0.1742$ ), with a per-cell dry-mass correlation of  $0.9363$  against  $0.8937$ . The field-total dry-mass MAPE was lower for the tested classical pipeline:  $0.1094$ , against  $0.1631 \pm 0.0141$  for the network (Sec. 5.2). All comparisons in this paragraph are with the tested classical pipeline only.

Table 5 compares the two geometries on the 107 test fields whose in-line hologram is correctly paired. The In-Line Neural Configuration reaches phase  $r = 0.8667$ , Dice  $0.8148$ , boundary F1  $0.4165$ , recall  $0.5885$  and a dry-mass MAPE of  $0.2129$  over 1798 matched cells. On the same fields the off-axis baseline has phase  $r = 0.8745 \pm 0.0023$ , Dice  $0.8330 \pm 0.0013$ , boundary F1  $0.4706 \pm 0.0052$ , recall  $0.5936 \pm 0.0031$  and dry-mass MAPE  $0.1740 \pm 0.0045$ . The in-line configuration was trained once, so these differences (for example  $+0.0389$  in dry-mass MAPE) are not resolvable under our criterion. The tested classical in-line reconstruction recovered near-background specimen contrast ( $r = -0.2075$ ), consistent with the behaviour reported for a 500-iteration Gerchberg–Saxton reconstruction in Park *et al.* [13] (Supplementary Section S2), whose phase profile remained near the background. Our in-line reconstruction used 20 iterations at  $33.77 \mu\text{m}$ . Its in-cell phase contrast (median over the 107 fields of the mean phase inside the reference cells minus the mean outside, computed identically for all columns of Table 5) is  $-0.084$  rad, against  $+0.922$  rad for the classical off-axis reconstruction on the same fields; the reference phase gives  $+1.097$  rad, the In-Line Neural Configuration  $+0.887$  rad and the off-axis baseline  $+0.899 \pm 0.036$  rad. The classical in-line pipeline matched 106 of 3055 reference cells (Dice  $0.1583$ ).

**Table 4:** End-to-End Neural Baseline and the tested classical pipeline on the off-axis test set (113 fields, 3186 reference cells). Neural values are mean  $\pm$  SD over three training seeds; the classical pipeline is deterministic and was run once.  $\Delta$ : neural minus classical.

| Quantity                                   | Classical | Neural ( $n = 3$ )  | $\Delta$            |
|--------------------------------------------|-----------|---------------------|---------------------|
| Phase MAE [rad] $\downarrow$               | 0.1968    | $0.1568 \pm 0.0023$ | $-0.0400$           |
| Phase MAE inside cells [rad] $\downarrow$  | 0.3112    | $0.2689 \pm 0.0056$ | $-0.0423$           |
| Phase Pearson $r$ $\uparrow$               | 0.7851    | $0.8741 \pm 0.0023$ | $+0.0890$           |
| Phase SSIM $\uparrow$                      | 0.5299    | $0.6386 \pm 0.0053$ | $+0.1087$           |
| Dice $\uparrow$                            | 0.8253    | $0.8337 \pm 0.0013$ | $+0.0085$           |
| AJI $\uparrow$                             | 0.6387    | $0.6710 \pm 0.0025$ | $+0.0322$           |
| Boundary F1 $\uparrow$                     | 0.3954    | $0.4687 \pm 0.0053$ | $+0.0733$           |
| Detection recall $\uparrow$                | 0.5869    | $0.5931 \pm 0.0027$ | $+0.0062$           |
| Detection precision $\uparrow$             | 0.7910    | $0.8488 \pm 0.0124$ | $+0.0577$           |
| Projected-area MAPE $\downarrow$           | 0.1578    | $0.1568 \pm 0.0027$ | $-0.0010$           |
| Dry-mass MAPE $\downarrow$                 | 0.2203    | $0.1742 \pm 0.0045$ | $-0.0461$           |
| Dry-mass Pearson $r$ , per cell $\uparrow$ | 0.8937    | $0.9363 \pm 0.0007$ | $+0.0427$           |
| Field-total dry-mass MAPE $\downarrow$     | 0.1094    | $0.1631 \pm 0.0141$ | $+0.0537$           |
| Cells matched (of 3186)                    | 1870      | $1890 \pm 9$        | $+20 \pm 9^\dagger$ |

$^\dagger$  Descriptive difference of the matched-cell count (neural mean minus classical; the SD is the between-seed SD of the neural count). A matched-cell count is not an error metric, and the resolution criterion of Sec. 4.8 is not applied to it. The classical pipeline is deterministic, so no difference in this table is assessed against a between-seed spread.



**Figure 2:** Neural and tested classical pipelines, oriented so that longer bars are better (error metrics drawn as  $1 - \text{error}$ , raw value printed). (a) Off-axis, 113 test fields; End-to-End Neural Baseline, mean over three seeds. (b) In-line, the 107 correctly paired test fields; In-Line Neural Configuration, one run. The negative phase correlation of the classical in-line reconstruction is drawn at zero with a left-pointing marker.

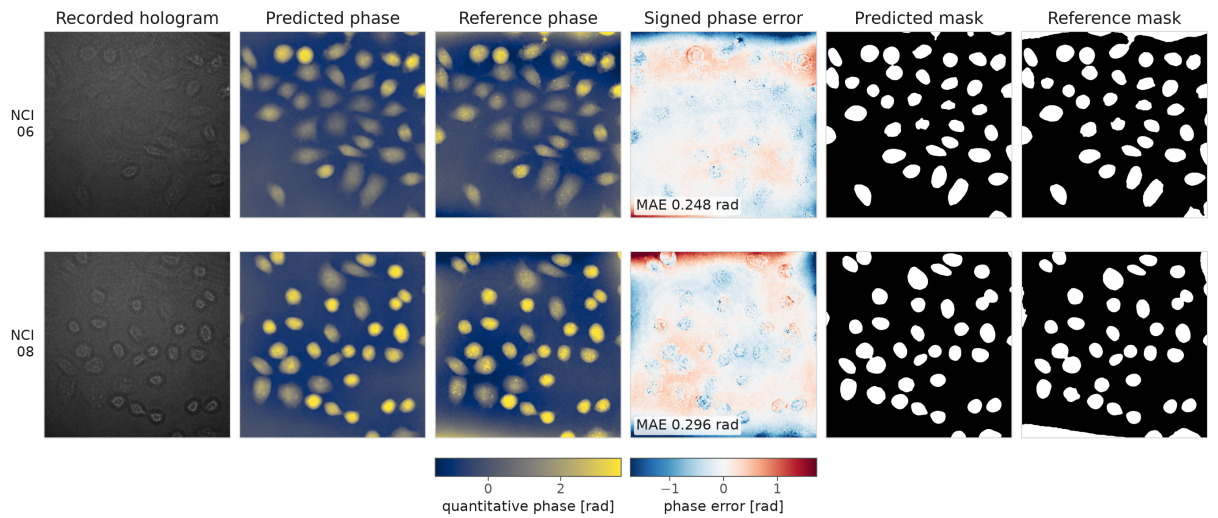
Its measurements are therefore computed from a phase map that does not carry the specimen contrast, and its per-cell dry-mass correlation, over 106 cells, is not comparable with the other entries. This result concerns the tested in-line configuration and does not describe classical in-line reconstruction in general.

**Table 5:** Off-axis and in-line pipelines on the 107 test fields common to both geometries (3055 reference cells; SNU\_01–SNU\_50 excluded). In-line neural: one training run; off-axis neural: mean  $\pm$  SD over three seeds, the same models as in Table 4 re-scored on these fields; classical pipelines: one deterministic run each.  $n$ : number of training runs.

| Quantity                                            | Neural              |                      | Classical |         |
|-----------------------------------------------------|---------------------|----------------------|-----------|---------|
|                                                     | In-line ( $n = 1$ ) | Off-axis ( $n = 3$ ) | Off-axis  | In-line |
| Recovered in-cell phase contrast [rad] <sup>‡</sup> | +0.887              | +0.899 $\pm$ 0.036   | +0.922    | −0.084  |
| Phase MAE [rad] ↓                                   | 0.1625              | 0.1586 $\pm$ 0.0024  | 0.1990    | 0.3996  |
| Phase MAE inside cells [rad] ↓                      | 0.3005              | 0.2726 $\pm$ 0.0057  | 0.3157    | 1.0448  |
| Phase Pearson $r$ ↑                                 | 0.8667              | 0.8745 $\pm$ 0.0023  | 0.7849    | −0.2075 |
| Phase SSIM ↑                                        | 0.6296              | 0.6377 $\pm$ 0.0053  | 0.5297    | 0.0639  |
| Dice ↑                                              | 0.8148              | 0.8330 $\pm$ 0.0013  | 0.8240    | 0.1583  |
| AJI ↑                                               | 0.6399              | 0.6697 $\pm$ 0.0031  | 0.6355    | 0.0549  |
| Boundary F1 ↑                                       | 0.4165              | 0.4706 $\pm$ 0.0052  | 0.3943    | 0.1087  |
| Detection recall ↑                                  | 0.5885              | 0.5936 $\pm$ 0.0031  | 0.5849    | 0.0347  |
| Detection precision ↑                               | 0.8406              | 0.8487 $\pm$ 0.0140  | 0.7883    | 0.0077  |
| Projected-area MAPE ↓                               | 0.1669              | 0.1552 $\pm$ 0.0025  | 0.1576    | 0.5315  |
| Dry-mass MAPE ↓                                     | 0.2129              | 0.1740 $\pm$ 0.0045  | 0.2204    | 0.6942  |
| Field-total dry-mass MAPE ↓                         | 0.1356              | 0.1657 $\pm$ 0.0145  | 0.1101    | 0.6602  |
| Dry-mass Pearson $r$ , per cell ↑                   | 0.9029              | 0.9374 $\pm$ 0.0010  | 0.8940    | 0.8758* |
| Cells matched (of 3055)                             | 1798                | 1813 $\pm$ 10        | 1787      | 106     |

\*Computed over the 106 cells matched by the classical in-line pipeline; not comparable with the other columns. <sup>‡</sup>Median over the 107 fields of the mean phase inside the reference cell masks minus the mean phase outside them, computed identically for all four columns; the reference phase gives +1.097 rad. The differences between the single in-line run and the three off-axis runs are not resolvable under the criterion of Sec. 4.8.

Figure 3 shows the seed-42 baseline on the first two test fields of the split, NCI\_06 and NCI\_08, without selection. The signed phase error is small over most of the background and largest in bands along the field edges, and the reference masks include regions along the field boundary that the network does not segment.



**Figure 3:** Outputs of the seed-42 End-to-End Neural Baseline on test fields NCI\_06 (top) and NCI\_08 (bottom): recorded off-axis hologram (cropped to the phase grid), predicted and reference phase on a shared colour scale, signed phase error (prediction minus reference, field MAE printed), and predicted and reference masks.

## 5.2 Per-cell measurements, error decomposition and population coverage

Over  $1890 \pm 9$  matched cells (mean over three seeds; the seed-42 run shown in Fig. 4 has 1898), the baseline’s per-cell dry-mass MAPE is  $0.1742 \pm 0.0045$  (seed-42 bootstrap interval 0.1650–0.1849) with a per-cell Pearson  $r$  of 0.9363. Projected area has MAPE 0.1568 and  $r = 0.8986$ , and circularity has MAPE 0.0571 and  $r = 0.8861$  (Table 6a). The per-field Bland–Altman bias, computed on per-field medians and averaged over seeds, is  $-0.0372$  for dry mass (limits  $[-0.247, +0.172]$ ) and  $+0.0649$  for projected area ( $[-0.162, +0.292]$ ), so on matched cells area is slightly over-estimated and dry mass under-estimated (Table 6b, Fig. 4). Field totals, which include unmatched cells on both sides, are low for both quantities ( $-0.0901$  area,  $-0.1560$  dry mass), with a field-total dry-mass correlation of 0.9883.

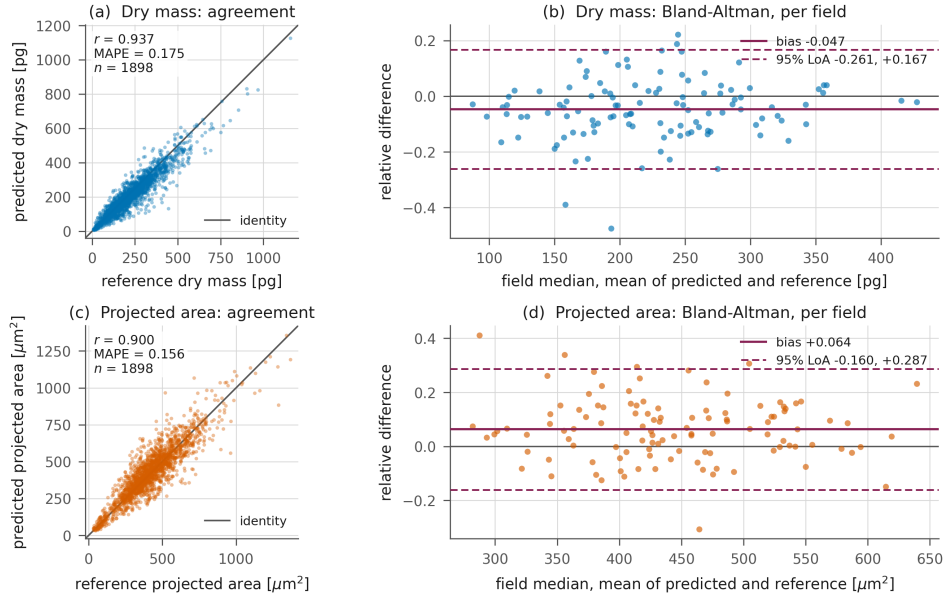
**Table 6:** Measurement agreement of the End-to-End Neural Baseline on the off-axis test set (113 fields, 3186 reference cells). Mean  $\pm$  SD over three training seeds unless stated. (a) Matched cells. (b) Relative Bland–Altman statistics on per-field medians of matched cells, and field totals over all predicted and all reference cells.

| (a) Matched cells |                     |                  |                     |                     |
|-------------------|---------------------|------------------|---------------------|---------------------|
| Quantity          | MAPE                | 95% CI (seed 42) | Cov.-adj. MAPE      | Pearson $r$         |
| Projected area    | $0.1568 \pm 0.0027$ | 0.1471–0.1663    | $0.4999 \pm 0.0036$ | $0.8986 \pm 0.0011$ |
| Circularity       | $0.0571 \pm 0.0027$ | 0.0493–0.0633    | $0.4408 \pm 0.0039$ | $0.8861 \pm 0.0072$ |
| Dry mass          | $0.1742 \pm 0.0045$ | 0.1650–0.1849    | $0.5102 \pm 0.0041$ | $0.9363 \pm 0.0007$ |

| (b) Per field and field totals |                          |                     |                      |                     |                     |
|--------------------------------|--------------------------|---------------------|----------------------|---------------------|---------------------|
| Quantity                       | Bland–Altman (per field) |                     | Field total          |                     |                     |
|                                | Bias                     | Limits of agreement | Bias                 | MAPE                | Pearson $r$         |
| Projected area                 | $+0.0649 \pm 0.0028$     | $[-0.162, +0.292]$  | $-0.0901 \pm 0.0037$ | $0.1553 \pm 0.0047$ | $0.9341 \pm 0.0043$ |
| Circularity                    | $+0.0217 \pm 0.0057$     | $[-0.054, +0.097]$  | N/A                  | N/A                 | N/A                 |
| Dry mass                       | $-0.0372 \pm 0.0172$     | $[-0.247, +0.172]$  | $-0.1560 \pm 0.0174$ | $0.1631 \pm 0.0141$ | $0.9883 \pm 0.0007$ |

CI: percentile bootstrap over fields (2000 resamples) of the seed-42 run. Limits of agreement: means over seeds. Field totals are not defined for circularity, which is non-additive. Dry mass and integrated phase  $S_k$  have identical relative statistics (Sec. 4.5).



**Figure 4:** Per-cell agreement of the seed-42 End-to-End Neural Baseline on the 113 off-axis test fields (1898 matched cells) for dry mass (a, b) and projected area (c, d): predicted against reference values (a, c) and relative Bland–Altman plots on per-field medians (b, d). The bias and limits of this run differ slightly from the three-seed means of Table 6.

Table 7 splits the predicted-to-reference dry-mass ratio with Eq. (17). For matched cells of the

baseline, the geometric-mean domain factor is  $1.002 \pm 0.002$  and the phase factor  $0.929 \pm 0.016$ , so the predicted domain encloses almost the same reference phase integral as the reference domain, and the systematic underestimate of matched cells lies in the predicted phase level. The phase factor also carries most of the spread (median  $|\log|$  0.102 against 0.042 for the domain factor). The phase factor is 0.957 for interior cells and 0.797 for edge cells. At field level the domain factor is 0.897; since the field-level domain includes every reference cell, this value reflects reference cells that were not detected, in addition to boundary placement. These statements concern matched cells; with a recall of 0.593, a matched-cell domain factor near 1 says nothing about the cells that were missed.

The tested classical pipeline has a different structure. For matched cells its domain factor is 0.966 and its phase factor 1.014, and at field level 0.857 and 1.055, giving a field-level total of 0.905 against 0.838 for the baseline. Opposing domain and phase deviations thus coincide with the lower field-total error of the classical pipeline reported in Sec. 5.1, which describes a compensating error structure. The factors and the field-total MAPE are different summaries, and the factors do not by themselves account for the MAPE values.

**Table 7:** Decomposition of the predicted-to-reference dry-mass ratio into a domain factor and a phase factor (Eq. (17)) on the off-axis test set (113 fields). GM: geometric mean of the factor (mean  $\pm$  SD over three seeds for the baseline);  $|\log|$ : median absolute log ratio (spread). Edge: reference instance with a pixel within 3 px of the field boundary.

| Level / group                             | N          | Recall | GM            |               |               | log    |       |       |
|-------------------------------------------|------------|--------|---------------|---------------|---------------|--------|-------|-------|
|                                           |            |        | Domain        | Phase         | Total         | Domain | Phase | Total |
| <i>End-to-End Neural Baseline (n = 3)</i> |            |        |               |               |               |        |       |       |
| Matched cells, all                        | 1889       | 0.593  | 1.002 ± 0.002 | 0.929 ± 0.016 | 0.931 ± 0.014 | 0.042  | 0.102 | 0.132 |
| Matched cells, interior                   | 1581       | 0.912  | 1.006 ± 0.002 | 0.957 ± 0.017 | 0.963 ± 0.015 | 0.038  | 0.090 | 0.116 |
| Matched cells, edge                       | 308        | 0.212  | 0.982 ± 0.004 | 0.797 ± 0.014 | 0.782 ± 0.014 | 0.076  | 0.267 | 0.303 |
| Field (foreground)                        | 113 fields | N/A    | 0.897 ± 0.002 | 0.933 ± 0.021 | 0.838 ± 0.017 | 0.105  | 0.073 | 0.179 |
| <i>Classical pipeline (n = 1)</i>         |            |        |               |               |               |        |       |       |
| Matched cells, all                        | 1870       | 0.587  | 0.966         | 1.014         | 0.980         | 0.046  | 0.119 | 0.169 |
| Matched cells, interior                   | 1543       | 0.890  | 0.977         | 1.028         | 1.004         | 0.041  | 0.106 | 0.152 |
| Matched cells, edge                       | 327        | 0.225  | 0.919         | 0.952         | 0.875         | 0.095  | 0.212 | 0.308 |
| Field (foreground)                        | 113 fields | N/A    | 0.857         | 1.055         | 0.905         | 0.141  | 0.061 | 0.095 |

N: matched cells per run (baseline: mean over seeds) or fields;  $n$  in the group labels is the number of training runs. Matched cells are recomputed by the decomposition script and differ from the stored evaluation by at most three cells per run; the re-scoring gives a per-run mean of 1889 matched cells, against  $1890 \pm 9$  from the stored evaluation (Table 4). Field level: the whole predicted and reference foreground, so undetected and spurious cells enter the domain factor.

The matched-cell errors describe the detected cells only. Detection recall is  $0.5931 \pm 0.0027$ , so about 1300 of the 3186 reference cells are not matched, and the coverage-adjusted dry-mass error is 0.5102. Stratifying by location (Table 8) shows that the missed cells are concentrated at the field boundary. 1452 reference instances (45.6%) have a pixel within 3 px of the boundary. The baseline recalls 0.212 of these edge instances and 0.912 of the interior ones, so about 1140 of the roughly 1300 unmatched reference cells are edge instances. Its matched-cell dry-mass MAPE is 0.305 on edge cells and 0.149 on interior cells. Edge instances include cells cut by the field boundary, whose area and mass are physically incomplete, and elongated boundary regions of the kind visible in Fig. 3; they were not classified individually.

Scored against the provisional membrane-derived masks instead of the phase-derived ones, the baseline’s Dice is 0.502 (against 0.834), AJI 0.236 (0.671), boundary F1 0.064 (0.469) and recall 0.114 (0.593), means over three seeds with between-seed SD at most 0.002. The membrane masks were scored over the whole field, including regions outside the fluorescence illumination, and they are thresholded perimeter stains. These values do not replace the phase-derived ones, but they show that agreement with the phase-derived labels does not carry over to an independent delineation of the same cells.

**Table 8:** Detection recall, matched-cell dry-mass MAPE and false positives by location of the reference instance on the off-axis test set (113 fields; 1452 edge and 1734 interior reference instances). Neural configurations: mean  $\pm$  SD over three seeds.

| Configuration                  | Recall            |                   | Dry-mass MAPE     |                   | False positives per run |              |
|--------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------------|--------------|
|                                | Edge              | Interior          | Edge              | Interior          | All                     | Within 15 px |
| End-to-End Neural Baseline     | $0.212 \pm 0.003$ | $0.912 \pm 0.002$ | $0.305 \pm 0.010$ | $0.149 \pm 0.004$ | $337 \pm 32$            | $136 \pm 16$ |
| +IPP (per-cell)                | $0.220 \pm 0.005$ | $0.919 \pm 0.002$ | $0.315 \pm 0.012$ | $0.163 \pm 0.003$ | $456 \pm 40$            | $203 \pm 22$ |
| +IPP (image)                   | $0.208 \pm 0.000$ | $0.907 \pm 0.003$ | $0.299 \pm 0.009$ | $0.167 \pm 0.003$ | $342 \pm 27$            | $136 \pm 14$ |
| Classical pipeline ( $n = 1$ ) | 0.225             | 0.890             | 0.322             | 0.199             | 494                     | 137          |

Edge: reference instance with a pixel within 3 px of the field boundary. Within 15 px: false-positive centroid within 15 px of the field boundary. Recall and MAPE are recomputed by the decomposition script (matched cells within three of the stored evaluation per run).

### 5.3 Measurement-aware training objectives

Table 9 lists, for every configuration, the matched-cell dry-mass MAPE next to the coverage-adjusted and field-total errors and the intermediate metrics; Table 10 gives the between-seed statistics of the replicated configurations and every dry-mass difference with its assessment under the resolution criterion. Figure 5 plots dry-mass MAPE and Dice for the configurations outside the weight sweep.

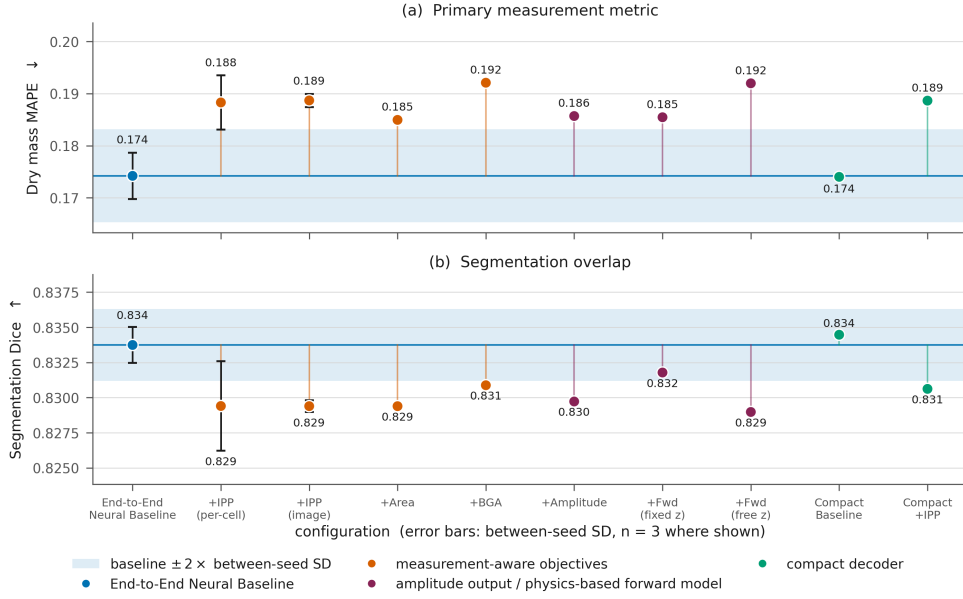
**Table 9:** Ablation over objective terms and decoder capacity on the off-axis test set (113 fields, 3186 reference cells).  $n$ : training runs;  $n = 3$  entries are means over seeds 42, 1337 and 2024,  $n = 1$  entries are seed-42 runs.

| Configuration              | $n$ | Dry-mass MAPE $\downarrow$ |           |             | Area<br>MAPE $\downarrow$ | Recall<br>$\uparrow$ | Precision<br>$\uparrow$ | Dice<br>$\uparrow$ | Phase<br>MAE [rad] $\downarrow$ |
|----------------------------|-----|----------------------------|-----------|-------------|---------------------------|----------------------|-------------------------|--------------------|---------------------------------|
|                            |     | matched                    | cov.-adj. | field total |                           |                      |                         |                    |                                 |
| End-to-End Neural Baseline | 3   | 0.1742                     | 0.5102    | 0.1631      | 0.1568                    | 0.5931               | 0.8488                  | 0.8337             | 0.1568                          |
| +IPP (per-cell)            | 3   | 0.1883                     | 0.5125    | 0.1676      | 0.1679                    | 0.6006               | 0.8077                  | 0.8294             | 0.1656                          |
| +IPP (image)               | 3   | 0.1887                     | 0.5222    | 0.1722      | 0.1597                    | 0.5889               | 0.8460                  | 0.8294             | 0.1644                          |
| +Area                      | 1   | 0.1850                     | 0.5075    | 0.1696      | 0.1779                    | 0.6042               | 0.7958                  | 0.8294             | 0.1646                          |
| +BGA                       | 1   | 0.1921                     | 0.5182    | 0.1651      | 0.1652                    | 0.5964               | 0.8183                  | 0.8309             | 0.1654                          |
| +IPP (per-cell), $w = 0.1$ | 1   | 0.1754                     | 0.5158    | 0.1830      | 0.1504                    | 0.5873               | 0.8428                  | 0.8331             | 0.1567                          |
| +IPP (per-cell), $w = 0.3$ | 1   | 0.1881                     | 0.5196    | 0.1953      | 0.1564                    | 0.5917               | 0.8389                  | 0.8328             | 0.1620                          |
| +IPP (per-cell), $w = 1.0$ | 3   | 0.1883                     | 0.5125    | 0.1676      | 0.1679                    | 0.6006               | 0.8077                  | 0.8294             | 0.1656                          |
| +IPP (per-cell), $w = 3.0$ | 1   | 0.1963                     | 0.5197    | 0.1316      | 0.1729                    | 0.5976               | 0.7855                  | 0.8164             | 0.1813                          |
| +Amplitude                 | 1   | 0.1857                     | 0.5157    | 0.1528      | 0.1591                    | 0.5948               | 0.8282                  | 0.8297             | 0.1633                          |
| +Fwd (fixed $z$ )          | 1   | 0.1855                     | 0.5081    | 0.1573      | 0.1685                    | 0.6039               | 0.8297                  | 0.8318             | 0.1647                          |
| +Fwd (free $z$ )           | 1   | 0.1920                     | 0.5212    | 0.1599      | 0.1583                    | 0.5926               | 0.8248                  | 0.8290             | 0.1636                          |
| Compact Baseline           | 1   | 0.1740                     | 0.5123    | 0.1738      | 0.1508                    | 0.5904               | 0.8496                  | 0.8345             | 0.1562                          |
| Compact +IPP               | 1   | 0.1887                     | 0.5088    | 0.1539      | 0.1656                    | 0.6055               | 0.8174                  | 0.8306             | 0.1652                          |

Differences from a single run ( $n = 1$ ) are not resolvable under the between-seed criterion (Sec. 4.8). Field totals sum all predicted and all reference cells and are dominated by the coverage deficit: the field-total dry-mass bias is negative in every configuration (Table 10a for the replicated ones). The  $w = 1.0$  row repeats +IPP (per-cell); a separate seed-42 training of the identical configuration gave a matched-cell MAPE of 0.1872 (seed-42 run of +IPP (per-cell): 0.1875).

Both measurement-aware integrated-phase terms increased matched-cell dry-mass MAPE (+IPP (per-cell)  $\Delta = +0.0141$ ,  $2 \times \text{SD } 0.0097$ ; +IPP (image)  $\Delta = +0.0145$ ,  $2 \times \text{SD } 0.0066$ ; both resolved under the predefined between-seed criterion). The two forms differ by  $+0.0004$ , inside their  $2 \times \text{SD}$  of 0.0076. Their effect on field-total dry-mass MAPE was not resolved ( $+0.0045$ ,  $2 \times \text{SD } 0.0232$ ;  $+0.0091$ ,  $2 \times \text{SD } 0.0250$ ). Several intermediate metrics also moved in the unfavourable direction by resolved amounts. Under +IPP (per-cell) these were area MAPE, phase MAE over the field and inside cells, phase  $r$ , SSIM, AJI, boundary F1 and precision ( $0.8488 \rightarrow 0.8077$ ), with 119 more false-positive cells per run; under +IPP (image) they were phase MAE over the field and inside cells, phase  $r$ , SSIM, Dice, AJI and boundary F1. +IPP (per-cell) also raised recall by 0.0075 (resolved), and at field level its domain factor rose from 0.897 to 0.909 ( $+0.0118$ ,  $2 \times \text{SD } 0.0081$ , resolved), together with the additional false positives, 67 of

which per run have centroids within 15 px of the field boundary (Table 8).



**Figure 5:** (a) Matched-cell dry-mass MAPE and (b) Dice of the off-axis configurations on the 113 test fields. Error bars: between-seed SD for the three configurations with three runs; other points are single runs. Shaded band: baseline mean  $\pm 2 \times$  its between-seed SD. Vertical axes do not start at zero.

The decomposition did not localise the increase consistently. No change in the geometric-mean domain or phase factor was resolved for all matched cells under either term, and the only resolved change in spread over all matched cells was a  $+0.0025$  increase of the domain  $|\log|$  spread under +IPP (image). The changes that were resolved are mostly for +IPP (image) and point in different directions for interior and edge cells: its interior phase spread rose by 0.0177, while its edge phase spread fell by 0.0352 and its edge phase factor rose by 0.0374. Under +IPP (per-cell) the resolved changes are a higher edge domain factor ( $+0.0154$ ) and edge total ( $+0.0341$ ). By location, the matched-cell MAPE increase is resolved among interior cells for both terms ( $+0.0139$  and  $+0.0183$ ) and not among edge cells (Table 8). The matched-cell domain factor did not change, so the route allowed by the formulation of Sec. 3.5.2, a foreground receding inside the reference domain with a raised phase, was not observed.

+Area and +BGA, each added to +IPP (per-cell), give dry-mass MAPE 0.1850 and 0.1921. Across the weight sweep (Fig. 6) the MAPE is 0.1754, 0.1881, 0.1883 and 0.1963 for  $w = 0.1, 0.3, 1.0$  and  $3.0$ , and at  $w = 3.0$  Dice (0.8164), precision (0.7855) and phase MAE (0.1813 rad) are the worst of the sweep. +Amplitude, the two +Fwd configurations, the Compact Baseline and Compact +IPP give 0.1857, 0.1855, 0.1920, 0.1740 and 0.1887. All of these are single runs, so none of their differences is resolvable. A separate seed-42 training of the configuration identical to +IPP (per-cell) gave 0.1872, against 0.1875 for the seed-42 run of +IPP (per-cell).

#### 5.4 Amplitude prediction

The amplitude head produces a non-constant transmittance map (predicted SD 0.1207 for +Amplitude) that agrees with the reconstruction-derived amplitude reference at an MAE of 0.0651 and a Pearson  $r$  of 0.9047, against an MAE of 0.1469 for the unit-amplitude assumption  $A = 1$ , a ratio of 0.443 (Table 11a). The two +Fwd configurations give 0.0665 and 0.0651 (ratios 0.453 and 0.444). These configurations were each trained once, and this is agreement with the output of one classical reconstruction, not with a measured transmittance. Dry mass depends on the amplitude only indirectly, through the shared encoder and phase decoder and through  $\mathcal{L}_{\text{fwd}}$ . Adding the amplitude output to +IPP (per-cell) changed the dry-mass MAPE by  $-0.0026$ , a single-run difference that is not resolvable.

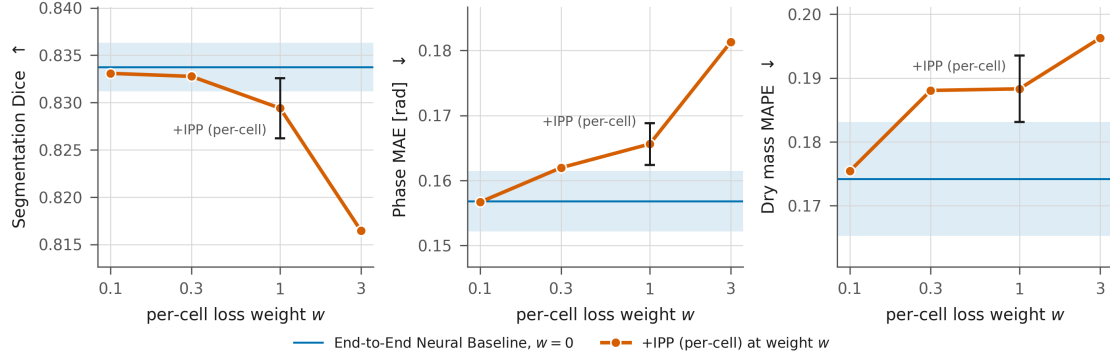
**Table 10:** Replicated configurations on the off-axis test set (113 fields). (a) Mean  $\pm$  SD over seeds 42, 1337 and 2024. (b) Differences in matched-cell and field-total dry-mass MAPE, configuration minus comparator, with  $2\times$  the pooled between-seed SD and the assessment.

| (a) Mean $\pm$ SD over three seeds |                      |                      |                      |
|------------------------------------|----------------------|----------------------|----------------------|
| Metric                             | Baseline             | +IPP (per-cell)      | +IPP (image)         |
| Dry-mass MAPE                      | 0.1742 $\pm$ 0.0045  | 0.1883 $\pm$ 0.0052  | 0.1887 $\pm$ 0.0013  |
| Field-total dry-mass MAPE          | 0.1631 $\pm$ 0.0141  | 0.1676 $\pm$ 0.0084  | 0.1722 $\pm$ 0.0107  |
| Field-total dry-mass bias          | -0.1560 $\pm$ 0.0174 | -0.1496 $\pm$ 0.0181 | -0.1667 $\pm$ 0.0078 |
| Area MAPE                          | 0.1568 $\pm$ 0.0027  | 0.1679 $\pm$ 0.0059  | 0.1597 $\pm$ 0.0014  |
| Phase MAE [rad]                    | 0.1568 $\pm$ 0.0023  | 0.1656 $\pm$ 0.0032  | 0.1644 $\pm$ 0.0015  |
| Phase MAE inside cells [rad]       | 0.2689 $\pm$ 0.0056  | 0.2876 $\pm$ 0.0064  | 0.2946 $\pm$ 0.0039  |
| Phase $r$                          | 0.8741 $\pm$ 0.0023  | 0.8636 $\pm$ 0.0019  | 0.8655 $\pm$ 0.0005  |
| Dice                               | 0.8337 $\pm$ 0.0013  | 0.8294 $\pm$ 0.0032  | 0.8294 $\pm$ 0.0004  |
| AJI                                | 0.6710 $\pm$ 0.0025  | 0.6616 $\pm$ 0.0059  | 0.6632 $\pm$ 0.0009  |
| Boundary F1                        | 0.4687 $\pm$ 0.0053  | 0.4518 $\pm$ 0.0093  | 0.4477 $\pm$ 0.0028  |
| Detection recall                   | 0.5931 $\pm$ 0.0027  | 0.6006 $\pm$ 0.0034  | 0.5889 $\pm$ 0.0013  |
| Detection precision                | 0.8488 $\pm$ 0.0124  | 0.8077 $\pm$ 0.0138  | 0.8460 $\pm$ 0.0100  |
| False positives per run            | 337 $\pm$ 32         | 456 $\pm$ 40         | 342 $\pm$ 27         |

| (b) Differences                                                                 |                |                     |                                       |
|---------------------------------------------------------------------------------|----------------|---------------------|---------------------------------------|
| Comparison                                                                      | $\Delta$       | $2\times$ pooled SD | Assessment                            |
| <i>Matched-cell dry-mass MAPE; both configurations with three training runs</i> |                |                     |                                       |
| +IPP (per-cell) vs Baseline                                                     | <b>+0.0141</b> | 0.0097              | Exceeds $2\times$ pooled SD           |
| +IPP (image) vs Baseline                                                        | <b>+0.0145</b> | 0.0066              | Exceeds $2\times$ pooled SD           |
| +IPP (image) vs +IPP (per-cell)                                                 | +0.0004        | 0.0076              | Within $2\times$ pooled SD            |
| <i>At least one configuration with a single training run</i>                    |                |                     |                                       |
| +Area vs +IPP (per-cell)                                                        | -0.0033        | N/A                 | Not estimable from the available runs |
| +BGA vs +IPP (per-cell)                                                         | +0.0038        | N/A                 | Not estimable from the available runs |
| $w = 0.1$ vs Baseline                                                           | +0.0012        | N/A                 | Not estimable from the available runs |
| $w = 0.3$ vs Baseline                                                           | +0.0138        | N/A                 | Not estimable from the available runs |
| $w = 3.0$ vs Baseline                                                           | +0.0221        | N/A                 | Not estimable from the available runs |
| +Amplitude vs +IPP (per-cell)                                                   | -0.0026        | N/A                 | Not estimable from the available runs |
| +Fwd (fixed $z$ ) vs +Amplitude                                                 | -0.0002        | N/A                 | Not estimable from the available runs |
| +Fwd (free $z$ ) vs +Fwd (fixed $z$ )                                           | +0.0065        | N/A                 | Not estimable from the available runs |
| Compact Baseline vs Baseline                                                    | -0.0002        | N/A                 | Not estimable from the available runs |
| Compact +IPP vs +IPP (per-cell)                                                 | +0.0003        | N/A                 | Not estimable from the available runs |
| <i>Field-total dry-mass MAPE; both configurations with three training runs</i>  |                |                     |                                       |
| +IPP (per-cell) vs Baseline                                                     | +0.0045        | 0.0232              | Within $2\times$ pooled SD            |
| +IPP (image) vs Baseline                                                        | +0.0091        | 0.0250              | Within $2\times$ pooled SD            |
| +IPP (image) vs +IPP (per-cell)                                                 | +0.0046        | 0.0193              | Within $2\times$ pooled SD            |
| <i>At least one configuration with a single training run</i>                    |                |                     |                                       |
| +Area vs +IPP (per-cell)                                                        | +0.0020        | N/A                 | Not estimable from the available runs |
| +BGA vs +IPP (per-cell)                                                         | -0.0025        | N/A                 | Not estimable from the available runs |
| $w = 0.1$ vs Baseline                                                           | +0.0199        | N/A                 | Not estimable from the available runs |
| $w = 0.3$ vs Baseline                                                           | +0.0322        | N/A                 | Not estimable from the available runs |
| $w = 3.0$ vs Baseline                                                           | -0.0315        | N/A                 | Not estimable from the available runs |
| +Amplitude vs +IPP (per-cell)                                                   | -0.0148        | N/A                 | Not estimable from the available runs |
| +Fwd (fixed $z$ ) vs +Amplitude                                                 | +0.0046        | N/A                 | Not estimable from the available runs |
| +Fwd (free $z$ ) vs +Fwd (fixed $z$ )                                           | +0.0025        | N/A                 | Not estimable from the available runs |
| Compact Baseline vs Baseline                                                    | +0.0107        | N/A                 | Not estimable from the available runs |
| Compact +IPP vs +IPP (per-cell)                                                 | -0.0137        | N/A                 | Not estimable from the available runs |

$\Delta$ : configuration minus comparator; positive is worse. Bold:  $|\Delta| > 2\sqrt{(s_1^2 + s_2^2)}/2$  with  $s$  the between-seed SD of each configuration, both configurations having at least three training runs. Not estimable: fewer than three training runs on one side, so no between-seed SD exists; no other variance is substituted. Differences are computed from unrounded means.



**Figure 6:** Segmentation Dice (left), phase MAE (centre) and matched-cell dry-mass MAPE (right) on the 113 off-axis test fields as functions of the  $\mathcal{L}_{\text{IPP}}^{\text{cell}}$  weight. The  $w = 1.0$  point is +IPP (per-cell), mean  $\pm$  SD over three seeds; the others are single runs. Line and band: baseline mean and  $\pm 2 \times$  its between-seed SD.

**Table 11:** (a) Amplitude output against the reconstruction-derived amplitude reference (113 off-axis test fields; one run each). (b) Forward-model residual on the 113 off-axis test fields (Baseline and +IPP (per-cell): mean  $\pm$  SD over three seeds; others one run). (c) Change in the residual when the reference phase is scaled by 0.9 or 0.5 (margin: scaled minus reference; positive means the reference phase has the lower residual).

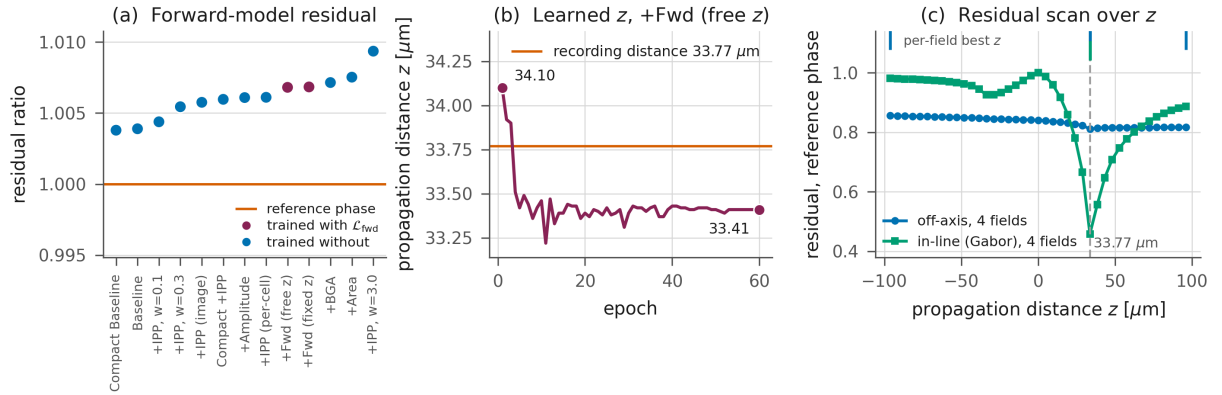
| (a) Amplitude output                           |                     |                     |                                  |                    |                                  |
|------------------------------------------------|---------------------|---------------------|----------------------------------|--------------------|----------------------------------|
| Quantity                                       | +Amplitude          |                     | +Fwd (fixed $z$ )                |                    | +Fwd (free $z$ )                 |
| MAE vs the reference                           | 0.0651              |                     | 0.0665                           |                    | 0.0651                           |
| the same MAE for $A = 1$                       | 0.1469              |                     | 0.1469                           |                    | 0.1469                           |
| ratio ( $< 1$ : closer than $A = 1$ )          | 0.4429              |                     | 0.4526                           |                    | 0.4436                           |
| MAE inside cells                               | 0.0906              |                     | 0.0922                           |                    | 0.0898                           |
| Bias                                           | +0.0339             |                     | +0.0378                          |                    | +0.0323                          |
| Pearson $r$                                    | 0.9047              |                     | 0.9064                           |                    | 0.9045                           |
| Predicted amplitude, mean                      | 1.0324              |                     | 1.0363                           |                    | 1.0308                           |
| Predicted amplitude, SD                        | 0.1207              |                     | 0.1223                           |                    | 0.1200                           |
| Reference amplitude, mean                      | 0.9985              |                     | 0.9985                           |                    | 0.9985                           |
| (b) Forward-model residual                     |                     |                     |                                  |                    |                                  |
| Quantity                                       | Baseline            | +IPP (per-cell)     | +Amplitude                       | +Fwd (fixed $z$ )  | +Fwd (free $z$ )                 |
| Residual, prediction                           | $0.8948 \pm 0.0002$ | $0.8968 \pm 0.0005$ | 0.8699                           | 0.8702             | 0.8699                           |
| Residual, reference phase                      | $0.8914 \pm 0.0000$ | $0.8914 \pm 0.0000$ | 0.8647                           | 0.8643             | 0.8640                           |
| Ratio (1.0 = reference phase)                  | $1.0039 \pm 0.0002$ | $1.0061 \pm 0.0005$ | 1.0061                           | 1.0068             | 1.0068                           |
| In-cell phase MAE [rad]                        | $0.2689 \pm 0.0056$ | $0.2876 \pm 0.0064$ | 0.2862                           | 0.2893             | 0.2886                           |
| (c) Phase-scale probes                         |                     |                     |                                  |                    |                                  |
| Geometry and fields                            | Residual            | 0.9 $\times$ phase  |                                  | 0.5 $\times$ phase |                                  |
|                                                |                     | Margin              | Fields favouring reference phase | Margin             | Fields favouring reference phase |
| Off-axis, 32 validation fields                 | 0.9147              | +0.00035            | 18/32                            | +0.0080            | 20/32                            |
| In-line, 32 validation fields                  | 0.6993              | +0.00432            | 30/32                            | +0.0622            | 32/32                            |
| Off-axis, 113 test fields, global surface      | 0.8914              | +0.00027            | 60/113                           | +0.0070            | 71/113                           |
| Off-axis, 112 test fields, per-field surfaces* | 0.3906              | +0.00110            | 93/112                           | +0.0359            | 110/112                          |

Fields favouring reference phase: number of fields on which the scaled phase gives the higher residual than the reference phase (a count of fields, not a separate test). Margin: scaled minus reference residual, averaged over the fields. Configured tolerance for a usable margin: 0.01. Residual levels in (b) for configurations with an amplitude output use the predicted amplitude in place of  $A = 1$  and are not comparable with the others; only the ratio is. \*Per-field aberration surfaces are fitted against each field's reference phase and are not available at inference. One test field (T24.Staurosporine\_100nM\_10) has no per-field surface: its fit was rejected because the fitted surface exceeded the 45 rad peak-to-valley limit (405 rad). With the global surface on the same 112 fields the residual is 0.8926, the margins are +0.00025 and +0.0067 and the counts 59/112 and 70/112.

## 5.5 Forward-model consistency

With the reference phase and the global aberration surface, the off-axis residual of Eq. (15) is 0.8914 on the 113 test fields (Table 11b): the forward model leaves most of the recorded hologram’s variance unexplained even for the reference phase. The network’s predictions come within 1% of that value. The ratio to the reference-phase residual lies between 1.0038 and 1.0094 for all 13 trained off-axis configurations (Fig. 7a),  $1.0039 \pm 0.0002$  for the baseline, which never optimised the term. +Fwd (fixed  $z$ ) and +Fwd (free  $z$ ) have ratios of 1.0068 and 1.0068, against 1.0061 for +Amplitude, and in-cell phase MAE of 0.2893 and 0.2886 rad against 0.2689 for the baseline; these are single runs. For the In-Line Neural Configuration (one run, 107 fields) the residual of the prediction is 0.7573 against 0.7654 for the reference phase.

Direct probes of the operator give the same picture (Table 11c). On 32 validation fields at the recording distance, the reference phase leaves a residual of 0.915 off-axis and 0.699 in-line. Scaling the reference phase by 0.9 raises the residual by 0.00035 off-axis (higher on 18 of the 32 fields) and by 0.0043 in-line (30 of 32); scaling it by 0.5 raises it by 0.0080 off-axis (20 of 32) and by 0.0622 in-line (32 of 32). Only the in-line  $0.5\times$  margin exceeds the configured tolerance of 0.01. On the 113 off-axis test fields, the  $0.9\times$  and  $0.5\times$  rescalings change the residual by  $+0.00027$  and  $+0.0070$  with the global surface (higher than the reference-phase residual on 60 and 71 of the 113 fields); with per-field surfaces, which are fitted against each field’s reference phase and are not available at inference and exist for 112 of the 113 fields (Table 11c), the residual is 0.391 and the changes are  $+0.0011$  and  $+0.0359$  (93 and 110 of the 112 fields). The per-field best-fitting phase scale has a median of 0.975 (IQR 0.575) off-axis and 1.00 (IQR 0.05) in-line. Under the implemented operator the residual therefore did not resolve the phase scale near the reference in either geometry.



**Figure 7:** Forward-model consistency. (a) Residual of each trained off-axis configuration on the 113 test fields divided by the residual obtained with the reference phase (three-seed means for the baseline and the two +IPP configurations, single runs otherwise). (b) Propagation distance of +Fwd (free  $z$ ) per epoch, initialised at the recording distance. (c) Residual computed from the reference phase as a function of  $z$  on four validation fields per geometry ( $4.81 \mu\text{m}$  grid); ticks at the top mark each field’s best distance, and the dashed line the recording distance.

With  $z$  trainable and initialised at  $33.77 \mu\text{m}$ , +Fwd (free  $z$ ) reached  $34.10 \mu\text{m}$  after the first epoch and  $33.41 \mu\text{m}$  after 60, with a total excursion of  $0.884 \mu\text{m}$  and a last-ten-epoch spread of  $0.018 \mu\text{m}$ ; the selected checkpoint (epoch 44) has  $z = 33.41 \mu\text{m}$  (Fig. 7b). A separate scan of the reference-phase residual over  $z$  on four validation fields (Fig. 7c) separates the two geometries. In-line, the residual falls from 1.000 at  $z = 0$  to a minimum of 0.458 at the grid point  $33.68 \mu\text{m}$ , the one nearest the recording distance, on all four fields (IQR  $0.0 \mu\text{m}$ ). Off-axis, the pooled residual also has its lowest value at  $33.68 \mu\text{m}$ , but the per-field best distances are  $33.68 \mu\text{m}$  on two fields and the two ends of the scanned range ( $\pm 96.2 \mu\text{m}$ ) on the other two (IQR  $48.1 \mu\text{m}$ ). Under the implemented operator the in-line residual

identifies the recording distance, and the off-axis residual did not constrain the propagation distance, so the learned value staying near its initialisation is not a measurement of the distance.

## 5.6 Sensitivity of quantitative measurements to segmentation boundaries

The analyses in this section use the reference phase and reference masks only (Table 12a, Fig. 8a,b). Displacing every reference boundary outward by one pixel ( $0.285\mu\text{m}$ ) changes projected area by  $+6.9\%$  and dry mass by  $+4.8\%$  while Dice is 0.974; an inward pixel gives  $-6.5\%$  and  $-4.7\%$ . At five pixels outward the changes are  $+32.6\%$  and  $+14.2\%$  at Dice 0.881. The ratio of mass error to area error has a median of 0.710 and falls with outward displacement (0.703 at  $+1\text{ px}$ , 0.450 at  $+5\text{ px}$ ).

**Table 12:** (a) Change in the measurements produced by a uniform displacement of the reference boundaries (113 off-axis test fields, reference phase and masks only). (b) Error of the measurement chain on synthetic fields with analytic mass and area (mean  $\pm$  SD over three independently generated sets).

| (a) Boundary displacement |                         |        |            |            |                 |
|---------------------------|-------------------------|--------|------------|------------|-----------------|
| Shift [px]                | Shift [ $\mu\text{m}$ ] | Dice   | Area error | Mass error | Mass/area ratio |
| -5                        | -1.424                  | 0.8452 | -0.2748    | -0.1966    | 0.717           |
| -4                        | -1.139                  | 0.8831 | -0.2269    | -0.1651    | 0.729           |
| -3                        | -0.855                  | 0.9127 | -0.1818    | -0.1332    | 0.733           |
| -2                        | -0.570                  | 0.9453 | -0.1229    | -0.0902    | 0.747           |
| -1                        | -0.285                  | 0.9731 | -0.0649    | -0.0474    | 0.735           |
| 0                         | +0.000                  | 1.0000 | +0.0000    | +0.0000    | -               |
| +1                        | +0.285                  | 0.9741 | +0.0693    | +0.0484    | 0.703           |
| +2                        | +0.570                  | 0.9493 | +0.1503    | +0.0964    | 0.678           |
| +3                        | +0.855                  | 0.9255 | +0.2138    | +0.1194    | 0.575           |
| +4                        | +1.139                  | 0.9056 | +0.2636    | +0.1317    | 0.518           |
| +5                        | +1.424                  | 0.8808 | +0.3262    | +0.1416    | 0.450           |

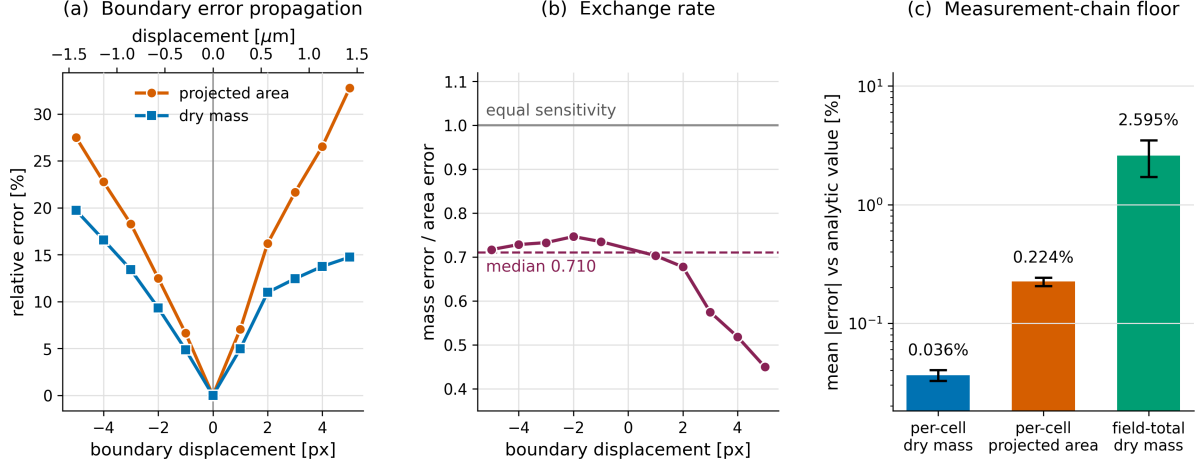
  

| (b) Measurement chain on analytic fields |                      |
|------------------------------------------|----------------------|
| Quantity                                 | Error                |
| Per-cell dry mass, mean  error           | $0.036 \pm 0.004\%$  |
| Per-cell projected area, mean  error     | $0.224 \pm 0.018\%$  |
| Field-total dry mass, mean bias          | $-2.592 \pm 0.889\%$ |

In the synthetic test (three sets of 12 fields with 12 separated spherical-cap cells each, radii  $2.5\text{--}8\mu\text{m}$ ), the mean absolute per-cell error of the measurement chain is  $0.036 \pm 0.004\%$  for dry mass and  $0.224 \pm 0.018\%$  for area. The field-total dry mass is biased by  $-2.59 \pm 0.89\%$ : cells with radius below  $3.09\mu\text{m}$  fall under the  $30\mu\text{m}^2$  area filter, and 10–17 of the 144 cells per set were not measured. These values describe the numerical error of the chain under this test condition, not the error to be expected on real cells, whose boundaries and profiles are not known exactly.

## 5.7 Computational benchmarking

At  $900 \times 900$  and batch 1 on an NVIDIA RTX A5000, the baseline runs at  $12.31 \pm 0.12\text{ ms}$  in PyTorch FP32 and  $8.42 \pm 0.07\text{ ms}$  in ONNX Runtime FP16, and the Compact Baseline, which has 65% fewer parameters, at  $7.97\text{ ms}$  in ONNX FP16 (Table 13, Fig. 9). Over all benchmark sessions the  $p_{99}/p_{50}$  latency ratio lies between 1.01 and 1.20. +IPP (per-cell), which shares the standard architecture, agrees with the baseline to within 0.12 ms except in PyTorch FP16, where its three sessions spread from 9.31 to 10.83 ms. The Compact Baseline changes the dry-mass MAPE by  $-0.0002$  relative to the baseline, inside the baseline’s  $2 \times \text{SD}$  of 0.0089; it was trained once, so the comparison is not resolvable. These are measurements on one workstation GPU.

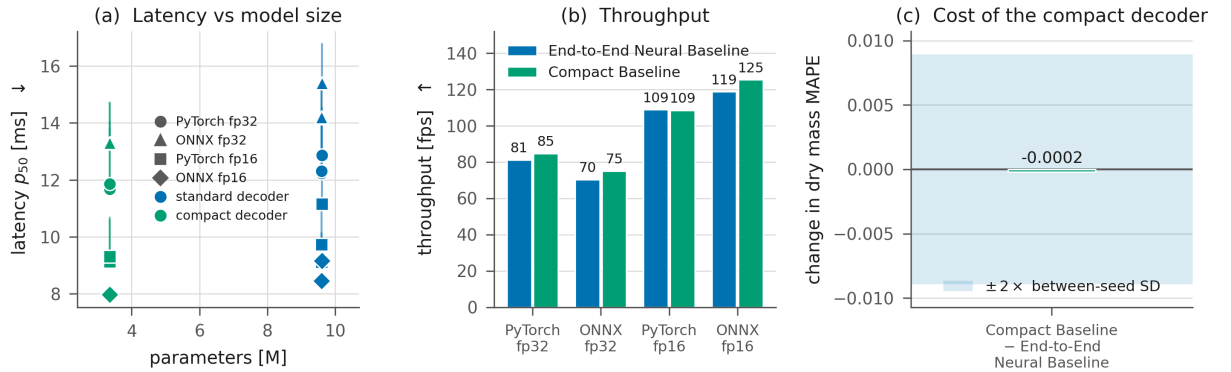


**Figure 8:** Model-free sensitivity analysis on the 113 off-axis test fields and synthetic fields: (a) area and dry-mass error under uniform boundary displacement of the reference masks, (b) ratio of dry-mass to area error, (c) error of the measurement chain on three synthetic sets (mean  $\pm$  SD; the field-total bar is the mean absolute error over fields).

**Table 13:** Computational benchmarking at  $900 \times 900$ , batch 1, on an NVIDIA RTX A5000. Baseline entries are mean  $\pm$  SD over three benchmark sessions, one per trained seed; the +Amplitude and Compact Baseline rows are each a single benchmark session.

| Configuration                                                     | Runtime      | Latency<br>[ms]  | $p_{99}/p_{50}$ | FPS             | Weights<br>[MB] | Peak alloc.<br>[MB] |
|-------------------------------------------------------------------|--------------|------------------|-----------------|-----------------|-----------------|---------------------|
| End-to-End Neural Baseline, $n = 3$<br>9 598 099 par., 45.85 GMAC | PyTorch FP32 | $12.31 \pm 0.12$ | $1.15 \pm 0.03$ | $81.2 \pm 0.8$  | 36.77           | 223.9               |
|                                                                   | ONNX FP32    | $14.21 \pm 0.08$ | $1.08 \pm 0.06$ | $70.4 \pm 0.4$  | —               | —                   |
|                                                                   | PyTorch FP16 | $9.19 \pm 0.10$  | $1.11 \pm 0.06$ | $108.9 \pm 1.2$ | 18.38           | 112.0               |
|                                                                   | ONNX FP16    | $8.42 \pm 0.07$  | $1.01 \pm 0.00$ | $118.8 \pm 0.9$ | —               | —                   |
| +Amplitude, $n = 1$<br>9 607 412 par., 47.87 GMAC                 | PyTorch FP32 | 12.97            | 1.13            | 77.1            | 36.81           | 223.9               |
|                                                                   | ONNX FP32    | 15.45            | 1.09            | 64.7            | —               | —                   |
|                                                                   | PyTorch FP16 | 11.26            | 1.12            | 88.8            | 18.40           | 111.4               |
|                                                                   | ONNX FP16    | 9.18             | 1.01            | 109.0           | —               | —                   |
| Compact Baseline, $n = 1$<br>3 360 403 par., 24.03 GMAC           | PyTorch FP32 | 11.80            | 1.20            | 84.7            | 12.97           | 195.7               |
|                                                                   | ONNX FP32    | 13.32            | 1.10            | 75.1            | —               | —                   |
|                                                                   | PyTorch FP16 | 9.21             | 1.16            | 108.5           | 6.48            | 98.7                |
|                                                                   | ONNX FP16    | 7.97             | 1.01            | 125.4           | —               | —                   |

Latency: mean over 500 timed passes after 50 warm-up passes. Weights and peak allocated memory are reported for PyTorch only.



**Figure 9:** Computational benchmarking on one RTX A5000: (a) median latency ( $p_{50}$ , whiskers to  $p_{99}$ ) against parameter count, (b) throughput by runtime and precision, (c) change in matched-cell dry-mass MAPE of the Compact Baseline relative to the End-to-End Neural Baseline, with the baseline's  $\pm 2 \times$  between-seed SD band (113 off-axis test fields).

## 6 Discussion

### 6.1 Measurement-oriented evaluation of the end-to-end pipeline

On this dataset one network that maps the raw off-axis hologram to phase and mask gives per-cell dry-mass estimates that agree with the reference measurements at a matched-cell MAPE of 0.174 and a per-cell correlation of 0.936, closer than the tested classical configuration (0.220 and 0.894). On the 107 correctly paired fields, the in-line configuration recovers phase and cell masks from holograms on which the tested classical in-line configuration recovered near-background specimen contrast, and its matched-cell dry-mass error (0.213) lies between those of the off-axis network and the off-axis classical pipeline; with one in-line run, the comparison with the off-axis network is not resolvable.

The closest systems in the literature reviewed in Sec. 2.4 predict dry mass from raw in-line holograms of plankton without a phase image [19], reconstruct phase and virtual fluorescence from Gabor holograms and compare measurement distributions [13], or chain reconstruction and detection without per-cell phase quantities [10]. What the present experiments add is a paired, per-cell account of the measurement for one joint network on experimental off-axis and in-line holograms, next to the fraction of reference cells the account covers and a split of the error into domain and phase. The comparison with those systems is qualitative; they use other specimens, other supervision and other instruments, and none of them was run on the data used here.

For learned holographic pipelines whose purpose is a per-cell quantity, the experiments point to three reporting practices. The per-cell error should be computed on matched cells and stated together with recall, because on this dataset the coverage-adjusted error (0.510) is almost three times the matched-cell error (0.174). Segmentation overlap should not stand in for measurement accuracy, because a Dice of 0.974 is compatible with a 5% mass error per cell. And the handling of cells at the field boundary should be part of the reference definition, because it decides a large part of the recall.

### 6.2 Why overlap metrics and measurements diverge

Against the tested classical pipeline, Dice (0.825 and 0.834) and projected area MAPE (0.158 and 0.157) are close, while the matched-cell dry-mass MAPE differs (0.220 and 0.174). The decomposition shows how the two errors are structured. For matched cells the network’s domains enclose almost the same reference phase integral as the reference domains (domain factor 1.002), and its error sits in the phase factor (0.929); the classical pipeline instead shows opposing domain and phase deviations (0.966 and 1.014). The lower in-cell phase MAE of the network (0.269 against 0.311 rad) is consistent with this split.

The model-free analysis gives the scale of the decoupling between overlap and measurement. A one-pixel boundary displacement leaves Dice at 0.974 but changes area by about 7% and dry mass by about 5%, so segmentation scores in the range reported here are compatible with measurement errors of ten percent or more. The observed matched-cell area error (0.157) is of the same order as the area change produced by a uniform two-pixel displacement ( $-12.3\%$  inward,  $+15.0\%$  outward). The displacement analysis applies a uniform shift and does not decompose the observed error, so this is a comparison of scale, not an attribution. Dry mass changes less than area under the same displacement (median ratio 0.71), which is consistent with the phase being low at the cell periphery. Classical DHM analyses report the same sensitivity from other directions: Bai *et al.* attribute part of an apparent volume increase to phase halos that place the segmented boundary outside the cell [6], and Kyeremah *et al.* describe morphological red-blood-cell features as sensitive to boundary precision [16]. The learned reconstructions reviewed in Sec. 2.1 are mostly scored with SSIM, PSNR or pixel error. By the numbers above, an image or overlap score in the range typically reported does not bound the per-cell measurement error.

### 6.3 Detection coverage and boundary objects

The matched-cell MAPE of 0.174 describes the roughly 1890 of 3186 reference cells that were detected and matched. With recall 0.593, the coverage-adjusted error is 0.510 and the field-total dry mass is low by about 16%, and the model also reports on average 337 false-positive cells per run, which enter the field totals but no per-cell error. The field-level domain factor (0.897) carries the same deficit. The location stratification shows that the missed cells are concentrated in one population: interior reference cells are recalled at 0.912 and edge instances at 0.212. Edge instances include cells truncated by the field of view and boundary regions that the reference masks retain because the border-removal step, as configured, has no effect after closing.

The field totals depend on how detection and measurement errors combine, and here the ordering of the two pipelines reverses: the classical pipeline has the higher matched-cell error but the lower field-total error (0.109 against 0.163). Its field-level domain and phase factors deviate in opposite directions (0.857 and 1.055), which describes a compensating error structure in which part of the domain deficit is offset by a raised phase. A field-total error alone would not separate this case from one in which both domain and phase are close to the reference.

Cells truncated by the field of view have physically incomplete area and dry mass, so whether they belong in a per-cell reference is a methodological choice. Two of the DHM studies we reviewed remove them before analysis: HoloPhaseNet discarded cells at the border of the phase image [9], and a SAM-based red-blood-cell pipeline removed cells touching the image edge [30]. Some of the low overall recall may therefore reflect objects that a different labelling rule would exclude. The edge instances were not classified individually, so the size of that share is not known. The stratification was not used to redefine any reported metric: recall and the coverage-adjusted error are those of the full reference population.

### 6.4 Measurement-aware objectives

Three observations are direct. Both integrated-phase terms increased the matched-cell dry-mass error by more than twice the pooled between-seed standard deviation, and neither changed the field-total error by a resolved amount. The matched-cell increase is resolved among interior cells and not among edge cells. And the per-cell term added about 119 false-positive cells per run and a resolved increase of the field-level domain factor, whereas the image-level term did not change the false-positive count.

One property of the formulation bears on the last observation.  $\mathcal{L}_{\text{IPP}}^{\text{cell}}$  sums only over reference instances, so predicted foreground outside them contributes nothing to it, whereas  $\mathcal{L}_{\text{IPP}}^{\text{img}}$  sums over the whole crop and so includes the phase under any false-positive foreground. The difference in false positives between the two terms is consistent with this, but the experiment does not isolate it as the cause. The formulation also admits a receding foreground with a raised phase as a way to satisfy  $\mathcal{L}_{\text{IPP}}^{\text{cell}}$  (Sec. 3.5.2); the unchanged matched-cell domain factor shows that the trained models did not take this route.

Two further possibilities were not tested. The first concerns the labels. The reference mask is a deterministic function of the reference phase, so when  $\mathcal{L}_{\text{phase}}$  and  $\mathcal{L}_{\text{seg}}$  are both well fitted, the quantity the integrated-phase terms constrain is already largely determined by them. An additional soft constraint on it may then add little information while adding gradient to the segmentation decoder, measured at initialisation as about 0.37 of the segmentation gradient at  $w = 1$ . If this is the mechanism, the terms could behave differently with segmentation labels that are independent of the phase target, such as masks derived from fluorescence acquired on the same field [18]. The second concerns the edge objects, which the per-cell term includes as integration domains during training. About half of the additional false positives under that term lie within 15 px of the field boundary, but the resolved increase in measurement error is among interior cells. Whether edge objects affect training in other ways would require retraining with regenerated masks.

The result is specific: these terms, at these weights, with these labels and this architecture, did not reduce the per-cell measurement error. It does not show that measurement-aware objectives cannot help, and it does not establish how the terms behave with other models or datasets. The two terms were compared at equal coefficients without matching their gradients. The +Area, +BGA, amplitude and forward-model configurations were trained once each, and single runs cannot exclude small effects in either direction. In the literature we reviewed, the only case where a measurement enters training is as a regression target learned from simulation [19], so these results cannot be set against a previous test of the same kind of term on experimental data.

### 6.5 What the forward-model term constrains

The forward-model term is the only component that relates the prediction to the recorded hologram through a model of image formation. Its behaviour differs between the geometries. In-line, the reference-phase residual has a clear minimum at the recording distance on every scanned field, so under the implemented operator the residual identifies  $z$ . Off-axis, it did not: the pooled minimum lies at the same grid point, but two of the four fields have their best distance at the ends of the scanned range. This is consistent with Eq. (3), in which the carrier encodes the phase at any distance, so the off-axis hologram changes little with  $z$ ; the scan does not test that explanation directly.

Near the reference phase, the residual is insensitive to the phase scale in both geometries. A 10% rescaling raises it by less than the configured tolerance of 0.01 off-axis (0.00035) and in-line (0.0043), although in-line the change has the same sign on 30 of the 32 fields. Off-axis, the global aberration surface leaves most of the variance unexplained even for the reference phase, and per-field surfaces, fitted against each field’s reference phase, leave much less: on the 112 test fields that have a per-field surface, the residual is 0.893 with the global surface and 0.391 with per-field surfaces. This is consistent with field-specific structure that the global surface does not represent accounting for part of the unexplained variance. With the term trained at  $w = 0.02$ , the +Fwd configurations did not reach a lower residual ratio than the baseline, and their in-cell phase error was higher; as single runs these differences are not resolvable.

The holographic forward-model methods discussed in Sec. 2.2 use in-line intensity with a distance that is fixed or obtained by autofocusing [26–28]. The free-distance configuration tested here treats the distance as an unknown calibration parameter of an off-axis model, the partially known case in the terminology of Ongie *et al.* [25]. These results characterise the implemented operator, evaluated on four fields per geometry for the distance scan and on 32 to 113 fields for the phase-sensitivity probes. They do not establish how other forward-model formulations would behave.

The amplitude output fits into the same picture. The network produces an amplitude map that agrees with the classical amplitude reference more closely than the unit-amplitude assumption, and this map completes the predicted complex field used by the forward model. Dry mass does not use the amplitude, so no direct effect on the measurement is expected; with single runs, none could be resolved.

## 7 Limitations

All segmentation, detection and matched-cell measurement results are relative to masks derived from the reference phase by a fixed thresholding rule. They do not measure agreement with an independent delineation, and the scores against the provisional membrane-derived masks are much lower. The Otsu level also varies with drug condition, so between-condition morphology comparisons made with these labels would partly reflect the label definition; no such comparison is reported here. As configured, the rule keeps truncated and boundary objects, and changing it would change the reference and require re-evaluating every configuration. The reference phase is itself a processed reconstruction whose exact processing is not known to us, and the amplitude reference is the modulus of a classical reconstruction;

neither is an independent physical measurement.

The hologram crop origin was determined empirically, on validation fields, by registration of a classical reconstruction to the reference phase. Its median residual on the test fields is below one pixel, but a per-field scatter of about  $\pm 2\text{--}3$  px (5–95% range) remains, so on some fields the network input and the phase target are offset by a few pixels.

The in-line result rests on a single training run, evaluated on 107 test fields, after excluding the 50 fields whose in-line frames do not correspond to their reference. The classical in-line reconstruction was run at one distance and with one algorithm.

Only the baseline and the two integrated-phase configurations have three runs. All other comparisons are single runs and are reported as not resolvable. The image-level term was run at the per-cell term’s coefficient without matching its gradient magnitude. The location stratification and the mass-error decomposition were carried out after training.

The forward-model conclusions apply to the implemented operator (scalar angular-spectrum propagation, a global order-5 aberration surface, a six-term radiometric fit off-axis) and to a term weight of 0.02. The distance scan covers four validation fields per geometry, and the per-field aberration surfaces used in one sensitivity probe are fitted against the reference phase.

All fields come from one instrument and three adherent cancer cell lines, and generalisation to other instruments, magnifications or cell types is untested. Computational cost was measured on one workstation GPU. The model outputs are predicted at half resolution and bilinearly upsampled. Absolute masses depend on  $\alpha = 0.2$  mL/g, the value used by the acquiring group, within a literature range of 0.173–0.215 mL/g [31, 32]; relative results do not.

## 8 Conclusions

A network that maps a raw off-axis hologram to quantitative phase and a cell segmentation measured per-cell dry mass on this dataset with a matched-cell MAPE of 0.174, against 0.220 for the tested classical configuration, at similar Dice and projected-area error. For matched cells the network’s error lies in the predicted phase level rather than in the predicted cell domain. For the full reference population, detection coverage is a larger limitation than the measurement error of detected cells (coverage-adjusted dry-mass error 0.510), and the missed cells are concentrated among boundary objects that the phase-derived labels retain. The field-total dry-mass MAPE was lower for the tested classical pipeline, and the decomposition describes a compensating error structure in it. On correctly paired in-line holograms, a single training run recovered phase where the tested classical in-line reconstruction did not.

Terms designed to preserve the integrated phase per cell or per image increased the matched-cell dry-mass error under the evaluated setup without a resolved change in the field-total error, and neither an area term, a boundary-alignment term, other weights, an amplitude output nor a forward-model term brought it below the baseline in single runs. Under the implemented operator, the forward-model residual identified the propagation distance in-line, did not constrain it off-axis, and did not resolve a 10% change of the phase scale in either geometry. A one-pixel boundary shift that leaves Dice at 0.974 changes dry mass by about 5%. Testing whether measurement-aware objectives help when segmentation labels are independent of the phase target, and whether a more complete forward model makes hologram consistency informative off-axis, are the direct next experiments.

## References

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